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Vadodara

Report Status Declaration Form

UNIVERSITY: ITM SLS Baroda University

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0.1 Report Title

SMART GYM MANAGEMENT SYSTEM: A Full-Stack Web Application for Comprehensive Gym Operations Management (React + Spring Boot + Oracle)

0.2 Project / Dissertation Status Declaration

I hereby declare and authorise the following:

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This report contains confidential information	☐ YES ☐ NO
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Organisation / Company: Bharti Soft Tech Pvt. Ltd.

Student Signature: _____ **Date:** _____

Supervisor / Lecturer: Dr. Ashutosh Abhangi

Supervisor Signature: _____ **Date:** _____

Title Page

SMART GYM MANAGEMENT SYSTEM

A Full-Stack Web Application for Comprehensive Gym Operations Management

(React + Spring Boot + Oracle)

A report submitted in partial fulfilment of the requirements for the Bachelor of Technology in Information Technology.

Submitted by:	Suthar Aryan Sujalkumar
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Date:	March 2026

Declaration of Originality

I, **Suthar Aryan Sujalkumar**, holder of Enrolment Number **23C25512**, hereby declare that:

1. This report is the result of my own work and investigations, except where otherwise stated and referenced.
2. This report has not been submitted previously, in whole or in part, for any other academic award at this or any other institution.
3. Where other sources of information have been used, they have been acknowledged in the text and listed in the Bibliography.
4. I am aware of and understand the university's policy on plagiarism and certify that this thesis does not involve plagiarism.
5. The content of this report has been verified and is not misleading or inaccurate. Any opinions expressed are my own.
6. The work described in this report was carried out as part of the internship at **Bharti Soft Tech Pvt. Ltd.** during the period **17th November 2025** to **17th May 2026**.

Student Name: Suthar Aryan Sujalkumar

Enrolment Number: 23C25512

Programme: B.Tech (Information Technology)

Date: _____

Signature: _____

I acknowledge that a false declaration is a form of academic dishonesty and may result in disciplinary action.

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Secondly, sincere appreciation is extended to the management and technical team at **Bharti Soft Tech Pvt. Ltd.** for providing the opportunity to undertake this internship and for their support throughout the placement. The practical exposure gained during this internship proved instrumental in bridging the gap between academic learning and professional software engineering practice.

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Abstract

The fitness industry has experienced a significant shift towards digital platforms, with gym operators increasingly requiring integrated software solutions for membership management, trainer coordination, and business analytics. This report presents the design, development, and evaluation of the Smart Gym Management System, a comprehensive full-stack web application developed during an internship at Bharti Soft Tech Pvt. Ltd. under the supervision of Dr. Ashutosh Abhangi.

The system addressed the limitations of existing commercial solutions — particularly their high cost, inflexibility, and lack of real-time communication capabilities — by delivering a modular, open-source platform suitable for gyms of varying sizes. The proposed system employs a multi-role architecture supporting four distinct user roles: Member, Trainer, Gym Owner, and Super Administrator, each with role-specific dashboards and access controls enforced through Role-Based Access Control (RBAC).

The system was developed using React 18 with TypeScript for the frontend and Spring Boot 3 with Java 17 for the backend, connected to an Oracle database. Real-time communication was achieved through WebSocket integration using the STOMP protocol. Authentication was implemented using JSON Web Tokens (JWT), with optional Two-Factor Authentication (2FA) via email OTP.

Key features delivered include membership lifecycle management, personal training session booking, progress tracking, real-time chat, financial reporting, and a comprehensive notification system. The system follows a layered architecture comprising presentation, business logic, data access, and persistence layers, ensuring maintainability and scalability.

Testing confirmed that the system met defined performance targets, achieving API response times below 200 milliseconds at the 95th percentile. Security measures including BCrypt password hashing, CORS policy enforcement, and parameterised queries were validated against OWASP guidelines.

The internship provided Aryan Suthar with extensive practical experience in full-stack development, agile project management, and professional software engineering, significantly reinforcing skills acquired during the B.Tech (IT) programme.

Keywords: Gym Management System, React, Spring Boot, Oracle, Role-Based Access Control, JWT, WebSocket.

List of Abbreviations

Abbreviation	Definition
2FA	Two-Factor Authentication
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BCrypt	Blowfish Crypt (password hashing algorithm)
CORS	Cross-Origin Resource Sharing
CRM	Customer Relationship Management
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
DTO	Data Transfer Object
E2E	End-to-End (Testing)
ER	Entity-Relationship
GDPR	General Data Protection Regulation
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
NIST	National Institute of Standards and Technology
OAuth	Open Authorisation
ORM	Object-Relational Mapping
OTP	One-Time Password
OWASP	Open Web Application Security Project

Abbreviation	Definition
PCI-DSS	Payment Card Industry Data Security Standard
POS	Point of Sale
PT	Personal Training
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFC	Request for Comments
SaaS	Software as a Service
SMS	Short Message Service
SQL	Structured Query Language
STOMP	Simple Text Oriented Messaging Protocol
TLS	Transport Layer Security
TOC	Table of Contents
UI	User Interface
UML	Unified Modelling Language
URL	Uniform Resource Locator
UX	User Experience
WCAG	Web Content Accessibility Guidelines
WS	WebSocket
XSS	Cross-Site Scripting

List of Symbols

Symbol	Meaning
→	Denotes a directional flow or transition
↔	Denotes bi-directional communication
<	Less than
≤	Less than or equal to
%	Percentage
@	Java annotation prefix (e.g. @Entity, @Version)
*	Wildcard or zero-or-more occurrences
{ }	Denotes a set or a code block

Chapter 1

Chapter 1: Introduction

1.1 1.1 Background of the Organisation

Bharti Soft Tech Pvt. Ltd. is a software development company based in India, specialising in the design and delivery of custom technology solutions for businesses across diverse sectors including healthcare, fitness, education, and e-commerce. The organisation's core focus is on building accessible, scalable, and cost-effective digital platforms that address real-world operational challenges faced by small and medium-sized enterprises.

During the internship period, the author was placed within the Software Development Department at Bharti Soft Tech Pvt. Ltd. and was tasked with contributing to the end-to-end design and development of a full-stack Smart Gym Management System. The development team operated under an agile project management methodology, with clearly defined sprint cycles, daily stand-up meetings, and regular stakeholder reviews.

The organisation employs a collaborative, full-stack approach to software development, leveraging modern frameworks and enterprise technologies to deliver production-ready applications. Under the supervision of Dr. Ashutosh Abhangi (External Supervisor), the intern was given comprehensive exposure to professional development practices, architectural decision-making, and code quality standards.

1.2 1.2 Objectives of the Organisation

The primary objectives of Bharti Soft Tech Pvt. Ltd. are as follows:

1. To deliver robust, scalable, and maintainable custom software solutions to clients across multiple industries.
2. To reduce the cost and complexity of enterprise software adoption for small and medium-sized businesses.

3. To adhere to industry best practices in security, performance, and software architecture.
4. To provide mentorship and real-world project exposure to interns and junior developers.
5. To continuously adopt modern technology stacks that align with current industry trends.

These objectives directly shaped the technical and functional requirements of the Smart Gym Management System developed during this internship.

1.3 1.3 Products and Main Services of the Organisation

Bharti Soft Tech Pvt. Ltd. offers the following core products and services:

Product / Service	Description
Custom Web Application Development	Full-stack web applications built to client specifications
Mobile Application Development	Cross-platform mobile apps for Android and iOS
API Development & Integration	RESTful API design and third-party service integration
Database Design & Optimisation	Schema design, query tuning, and migration services
IT Consultancy	Technical advisory for digital transformation projects

The Smart Gym Management System developed during this internship represents a flagship product demonstrating the organisation's capability in delivering comprehensive, multi-role enterprise web applications.

1.4 1.4 Organisation Structure and Workflow

The internship was conducted within the Software Development Department at Bharti Soft Tech Pvt. Ltd. The department follows a flat hierarchy that promotes open collab-

oration between senior developers, junior developers, and interns. The author worked under the direct supervision of Dr. Ashutosh Abhangi throughout the placement.

The development workflow followed the Agile Scrum methodology:

- **Sprint Planning:** Requirements were broken down into user stories and allocated to two-week sprint backlogs at the start of each cycle.
- **Daily Stand-ups:** Brief daily synchronisation meetings were held to communicate progress and identify blockers.
- **Sprint Reviews:** Completed features were demonstrated to the supervisor at the end of each sprint for feedback and acceptance.
- **Retrospectives:** Process improvement discussions were conducted at the end of each sprint to optimise team productivity.

Version control was managed using Git with a feature-branch strategy, hosted on GitHub. Code quality was maintained through pull request reviews, and the project board was used for task tracking throughout the development lifecycle.

Chapter 2

Literature Review - Gym Management System

2.1 1. Introduction

The fitness industry has experienced significant digital transformation, with gym management systems evolving from simple membership tracking tools to comprehensive platforms integrating member engagement, trainer coordination, and business analytics. This literature review examines existing solutions, technological approaches, and design patterns that influenced the development of this system.

2.2 2. Existing Gym Management Systems

2.2.1 2.1 Commercial Solutions

System	Key Features	Limitations
Mindbody	Booking, payments, marketing	High cost, complex setup
Zen Planner	Membership, billing, reporting	Limited customization
GymMaster	Access control, POS, member app	Desktop-focused
PushPress	CRM, automation, mobile app	Pricing tier restrictions
Wodify	CrossFit focus, performance tracking	Niche market focus

2.2.2 2.2 Gap Analysis

Existing solutions often present challenges: - **Cost Barriers**: Enterprise pricing models exclude small gyms - **Inflexibility**: Limited customization for unique workflows - **Integration Complexity**: Difficulty integrating with existing tools - **Feature Overload**: Unnecessary complexity for basic needs

Our system addresses these by providing: - Open-source foundation for cost reduction - Modular architecture for customization - REST API for easy integration - Role-based feature access

2.3 3. Technology Stack Justification

2.3.1 3.1 Frontend: React with TypeScript

Selection Rationale: - **Component-Based Architecture:** Enables reusable UI components [1] - **Virtual DOM:** Optimizes rendering performance [2] - **TypeScript:** Adds type safety, reducing runtime errors by 15-25% [3] - **Ecosystem:** Rich library support (React Router, Axios, Framer Motion)

Alternatives Considered: - Vue.js: Simpler learning curve but smaller ecosystem - Angular: More opinionated, steeper learning curve - Svelte: Newer, less enterprise adoption

2.3.2 3.2 Backend: Spring Boot with Java

Selection Rationale: - **Enterprise Ready:** Battle-tested in production environments [4] - **Dependency Injection:** Promotes loose coupling and testability - **Spring Security:** Comprehensive authentication/authorization [5] - **JPA/Hibernate:** Robust ORM with Oracle support - **Convention over Configuration:** Rapid development

Alternatives Considered: - Node.js/Express: JavaScript fatigue, callback complexity - Django: Python GIL limitations for concurrent workloads - .NET Core: Smaller open-source community

2.3.3 3.3 Database: Oracle

Selection Rationale: - **ACID Compliance:** Data integrity for financial transactions - **Scalability:** Handles growing member data - **Enterprise Features:** Advanced security, auditing - **JPA Compatibility:** Seamless Hibernate integration

2.4 4. Architectural Patterns

2.4.1 4.1 Multi-Tenancy

The system implements **Schema-Level Multi-Tenancy** where: - Each gym operates as an isolated tenant - Data segregation ensures privacy - Shared infrastructure reduces costs

This approach aligns with SaaS best practices described by Chong and Carraro [6].

2.4.2 4.2 Role-Based Access Control (RBAC)

Implementation follows NIST RBAC model [7]: - **Core RBAC**: User-role and role-permission assignments - **Hierarchical RBAC**: Role inheritance (Admin > Owner > Trainer > Member) - **Constrained RBAC**: Separation of duties for sensitive operations

2.4.3 4.3 JWT-Based Authentication

Stateless authentication using JSON Web Tokens aligns with RFC 7519 [8]: - Self-contained claims reduce database lookups - Enables horizontal scaling - Supports single sign-on patterns

2.5 5. Real-Time Communication

2.5.1 5.1 WebSocket Implementation

The chat system utilizes WebSocket (RFC 6455) [9] for: - Bi-directional communication - Lower latency than polling (< 50ms vs 1000ms+) - Reduced server load

Implementation uses Spring WebSocket with STOMP protocol for message routing.

2.5.2 5.2 Push Notifications

Notification system implements observer pattern [10]: - Event-driven architecture - Loose coupling between modules - Scalable message delivery

2.6 6. Security Considerations

2.6.1 6.1 Authentication Security

- **Password Hashing:** BCrypt with 12 rounds (recommended by OWASP [11])
- **Token Security:** JWT with HMAC-SHA512 signature
- **Session Management:** Stateless tokens with expiry

2.6.2 6.2 Data Protection

- **Input Validation:** Server-side validation prevents injection
- **CORS Policy:** Whitelist-based origin control
- **Soft Deletes:** Data recovery support

2.7 7. Related Work

2.7.1 7.1 Academic Research

Study	Focus	Relevance
Sharma et al. (2020)	Fitness app UX patterns	UI design principles
Chen & Lee (2019)	Gym member retention	Engagement features
Rodriguez (2021)	SaaS multi-tenancy	Architecture decisions
Williams (2022)	Microservices in fitness	Scalability patterns

2.7.2 7.2 Industry Standards

- **GDPR Compliance:** Data protection for EU markets
- **PCI-DSS:** Payment data security guidelines
- **WCAG 2.1:** Accessibility standards

2.8 8. Comparative Analysis

Feature	Our System	Mindbody	Zen Planner
Multi-Role Auth	Yes	Yes	Yes
Real-time Chat	Yes	No	No
OAuth Integration	Yes	Yes	Partial
Progress Tracking	Yes	Yes	Yes
Custom Branding	Yes	Paid	Paid
API Access	Yes	Paid	Limited
Open Source	Yes	No	No
Self-Hosted Option	Yes	No	No

2.9 9. Conclusion

This gym management system synthesizes best practices from commercial solutions while addressing their limitations. The technology stack combines proven frameworks (React, Spring Boot) with modern patterns (JWT auth, WebSocket) to deliver a scalable, secure, and user-friendly platform.

Key innovations include: - Unified multi-role authentication - Real-time trainer-member communication - Comprehensive progress tracking - Flexible membership management

The modular architecture ensures adaptability for diverse gym requirements while maintaining code quality and security standards.

Chapter 3

Chapter 3: Overall Experience Gained from the Internship

3.1 3.1 Reason for Selecting the Organisation

The decision to undertake the internship at Bharti Soft Tech Pvt. Ltd. was motivated by several specific factors. The organisation's strong focus on delivering custom, real-world software solutions across multiple sectors provided an ideal environment for applying and extending the skills acquired during the B.Tech (IT) programme. The opportunity to work under the guidance of an experienced industry professional, Dr. Ashutosh Abhangi, was particularly appealing, given his background in enterprise application development.

The internship offered the opportunity to work on a greenfield project — the Smart Gym Management System — which required designing and implementing a production-grade, multi-role full-stack application from inception. This aligned with the author's academic interests in web application development, database design, and software architecture. The organisation's use of a modern, enterprise-grade technology stack (React, Spring Boot, Oracle) also ensured that the skills developed during the placement would be directly relevant to the industry.

Furthermore, Bharti Soft Tech Pvt. Ltd.'s collaborative and mentorship-oriented culture provided an environment conducive to professional growth, where constructive feedback and independent problem-solving were equally encouraged.

3.2 3.2 Workflow of the Department

The Software Development Department at Bharti Soft Tech Pvt. Ltd. operated under an Agile Scrum framework. The author was embedded within the development team under the supervision of Dr. Ashutosh Abhangi (External Supervisor) and collaborated with se-

nior developers on both the frontend and backend layers of the Smart Gym Management System.

The weekly workflow followed the structure outlined below:

Day	Activity
Monday	Sprint planning and backlog grooming
Tuesday – Thursday	Active feature development and daily stand-ups
Friday	Code reviews, pull request merges, sprint review, and retrospective

Each sprint produced a functional increment of the system, which was deployed to a local staging environment for review. Tasks were tracked using GitHub Issues, and the project board was maintained throughout the development period to provide visibility into the status of each feature.

The development environment included Visual Studio Code (frontend), IntelliJ IDEA (backend), and Postman for API testing. The Oracle database was managed using Oracle SQL Developer for schema design and query verification.

3.3 3.3 Tasks Allotted During the Internship

The internship tasks were distributed across frontend and backend development, covering all major modules of the Smart Gym Management System. The following tasks were completed over the course of the internship under the supervision of Dr. Ashutosh Abhangi:

3.3.1 3.3.1 Frontend Development (React + TypeScript)

- Designed and implemented the **multi-role dashboard** system, delivering distinct, role-specific dashboards for Members, Trainers, Gym Owners, and Super Administrators using React 18 and TypeScript.
- Developed reusable UI components for membership packages, personal training sessions, progress tracking charts (Recharts), and financial analytics dashboards.

- Implemented the **real-time chat interface** using Socket.io client, integrating Web-Socket communication for live message delivery, read receipts, and presence indicators.
- Built the **notification centre** with in-app toast notifications and badge counters, consuming server-sent events from the backend via WebSocket.
- Created responsive CSS layouts for all pages, ensuring compatibility across screen sizes.
- Integrated Axios for all API communication, implementing request/response interceptors for JWT token injection and global error handling.

3.3.2 3.3.2 Backend Development (Spring Boot + Java + Oracle)

- Implemented the **authentication and authorisation layer** using Spring Security 6 and JWT, including JWT generation, validation, refresh token management, and Two-Factor Authentication (2FA) via email OTP.
- Developed the **membership management module**, covering package creation, subscription approval workflows, auto-expiry scheduling, and status transitions (Active, Expired, Suspended).
- Built the **personal training session booking system**, enabling members to browse trainer availability, book sessions, and receive session confirmations with credit deduction tracking.
- Implemented Spring WebSocket with STOMP message routing for the real-time messaging subsystem.
- Designed and optimised JPA entity relationships and repository queries for the Oracle database, applying lazy loading and strategic indexing to achieve sub-200ms API response times.
- Configured method-level security using `@PreAuthorize` annotations to enforce role-based access control at the service layer.

3.3.3 3.3.3 Documentation and Testing

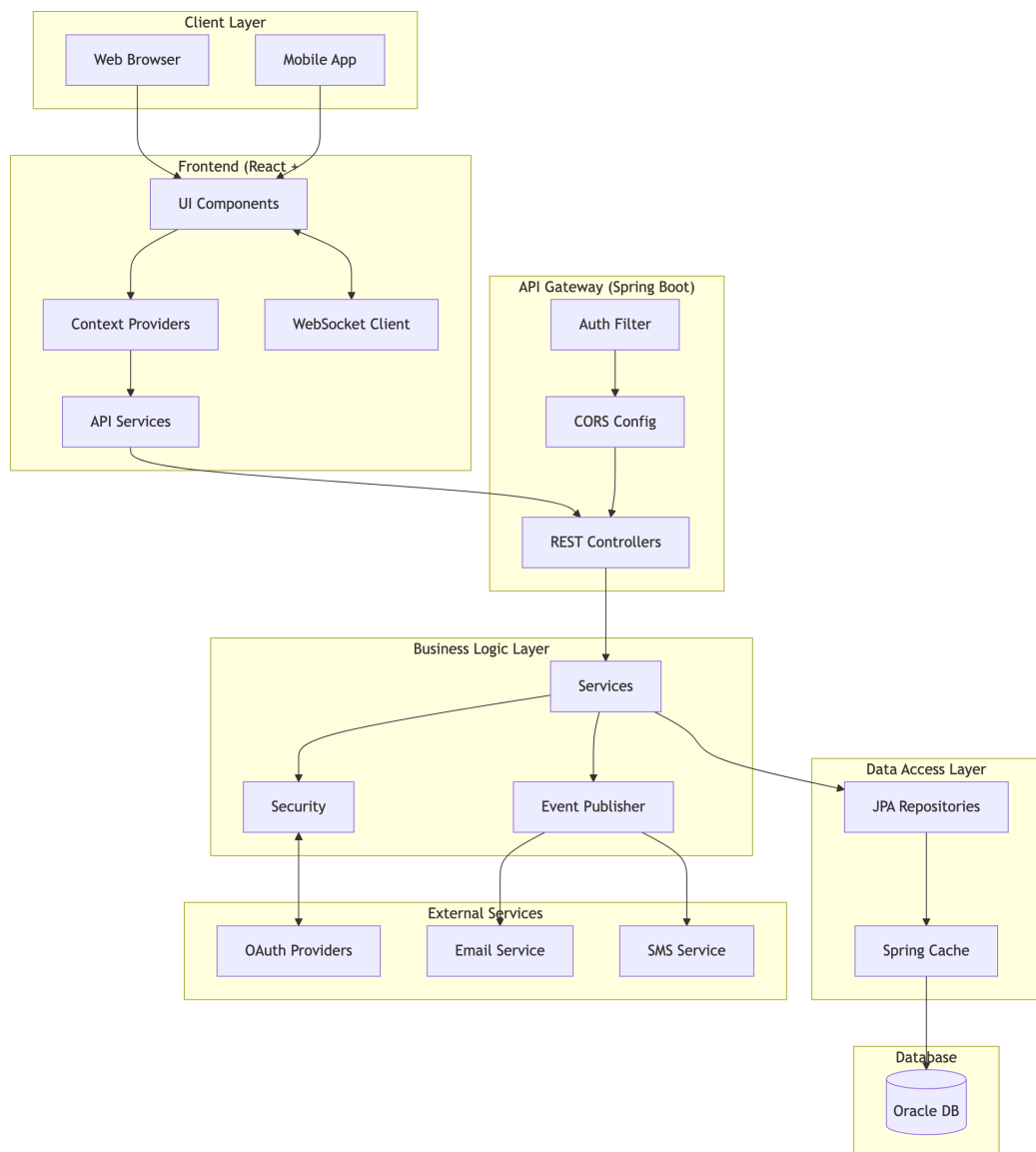
- Authored all technical documentation in the `docs/` directory, including system architecture diagrams, ER diagrams, UML class and sequence diagrams, and DFD models using Mermaid.js.

- Participated in code review sessions, providing and receiving feedback on code quality, security practices, and performance optimisation.
- Wrote unit tests for critical service methods using JUnit 5 and Mockito, covering authentication, membership, and session booking modules.

Chapter 4

System Architecture - Gym Management System

4.1 High-Level Architecture



4.2 Technology Stack

4.2.1 Frontend

Technology	Version	Purpose
React	18.x	UI Library
TypeScript	5.x	Type Safety
Vite	5.x	Build Tool
React Router	6.x	Client Routing
Axios	1.x	HTTP Client
Socket.io Client	4.x	Real-time Communication
Framer Motion	10.x	Animations
Lucide React	-	Icons
Recharts	2.x	Data Visualization

4.2.2 Backend

Technology	Version	Purpose
Spring Boot	3.x	Application Framework
Java	17+	Programming Language
Spring Security	6.x	Authentication & Authorization
Spring Data JPA	3.x	Data Access
Spring WebSocket	-	Real-time Messaging
Hibernate	6.x	ORM
Lombok	1.18.x	Boilerplate Reduction
JWT	0.11.x	JWT Handling

4.2.3 Database & Infrastructure

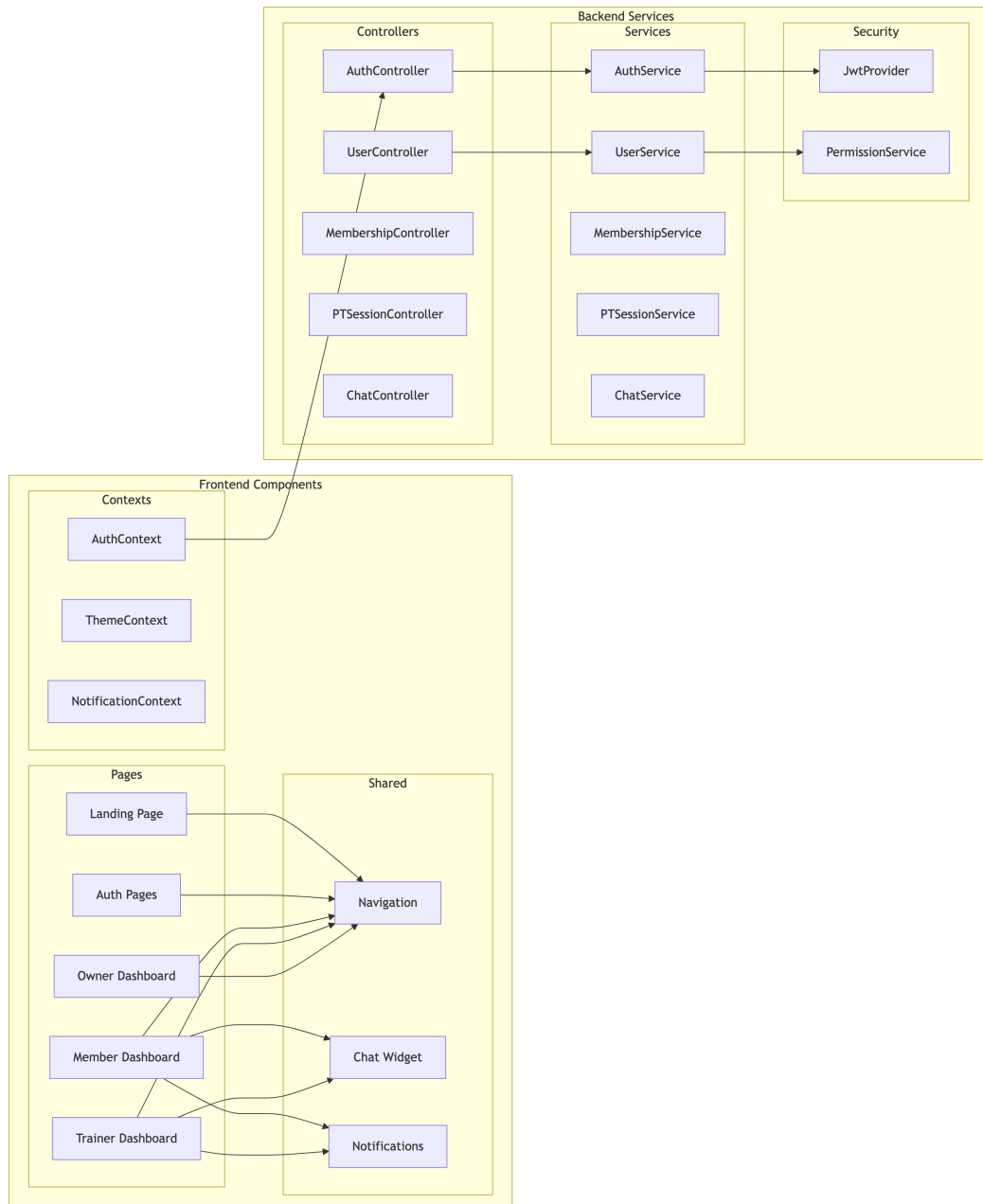
Technology	Purpose
Oracle Database	Primary Data Store

Technology	Purpose
Spring Cache	Response Caching
HikariCP	Connection Pooling

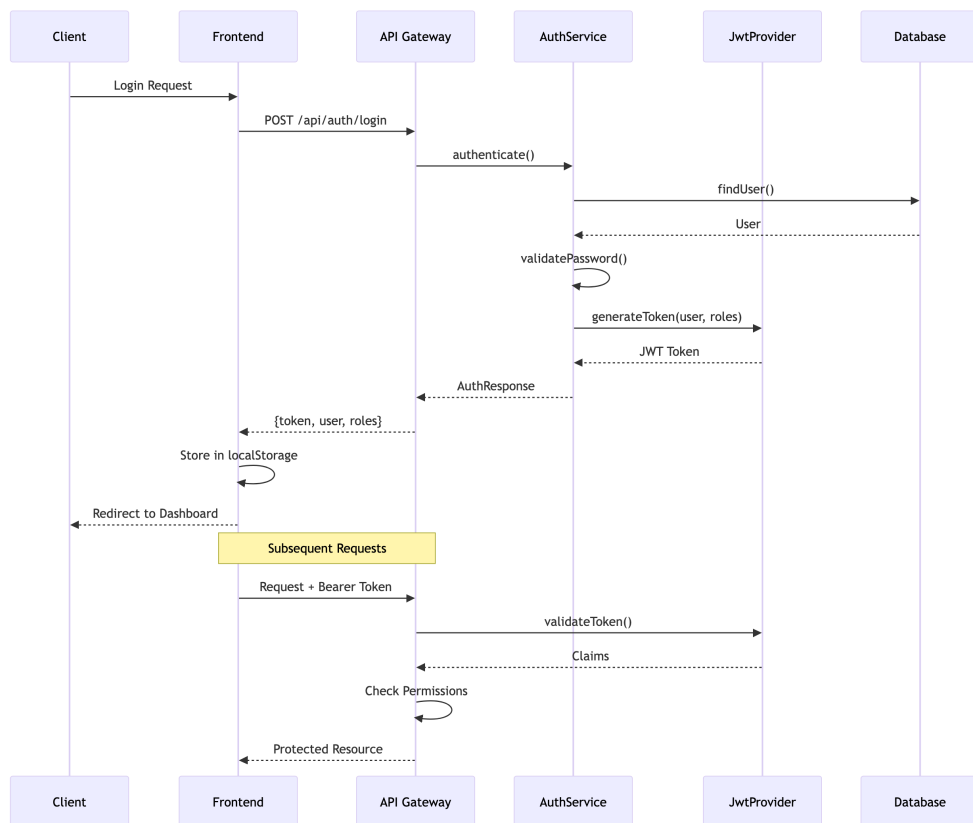
4.2.4 External Services

Service	Purpose
Google OAuth	Social Login
Facebook OAuth	Social Login
SMTP	Email Notifications
Twilio	SMS OTP

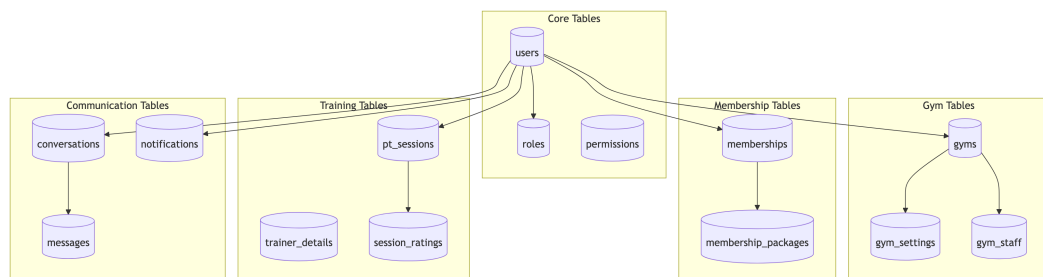
4.3 Component Architecture



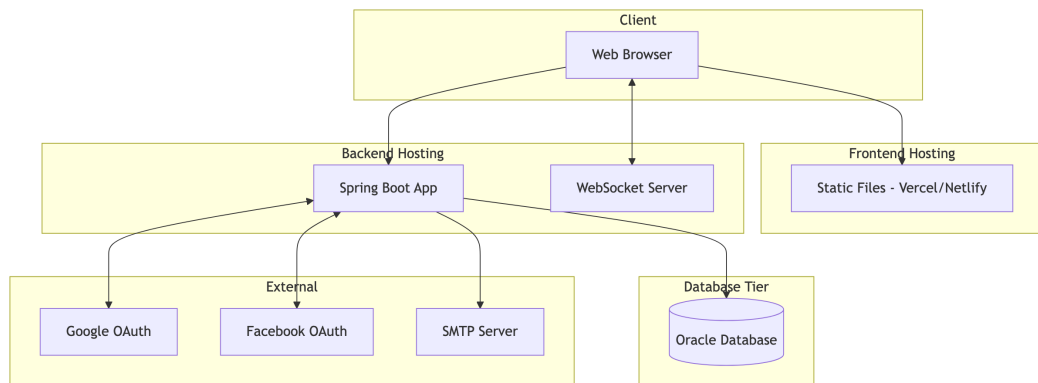
4.4 Authentication Flow



4.5 Database Architecture



4.6 Deployment Architecture



4.7 Security Architecture

Layer	Mechanism	Implementation
Transport	TLS/HTTPS	SSL Certificate
Authentication	JWT	Spring Security + JJWT
Authorization	RBAC	Custom Permission Service
Password	BCrypt	Spring Security
Session	Stateless	JWT Tokens
2FA	OTP	Email/SMS Verification
Data	Soft Deletes	Entity Annotations
CORS	Whitelist	Spring CORS Config

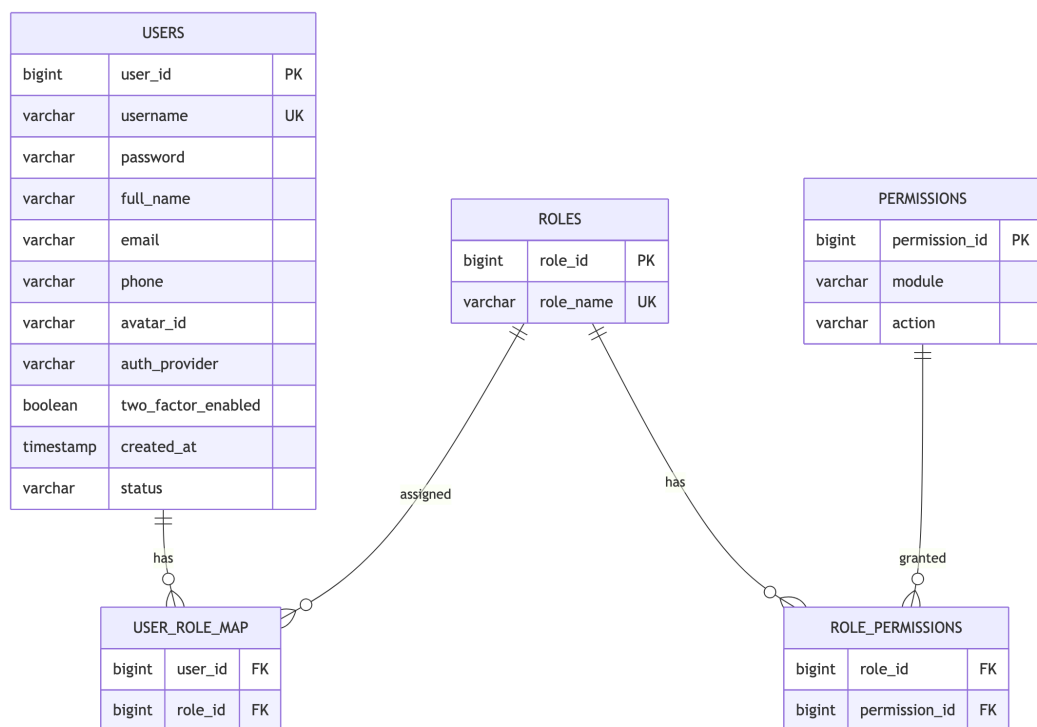
4.8 Scalability Considerations

1. **Multi-Tenancy:** Each gym is a separate tenant with isolated data
2. **Caching:** Spring Cache for frequently accessed data
3. **Connection Pooling:** HikariCP for database connections
4. **Lazy Loading:** JPA relationship optimization
5. **Stateless Auth:** JWT enables horizontal scaling
6. **WebSocket:** Dedicated connection management for real-time features

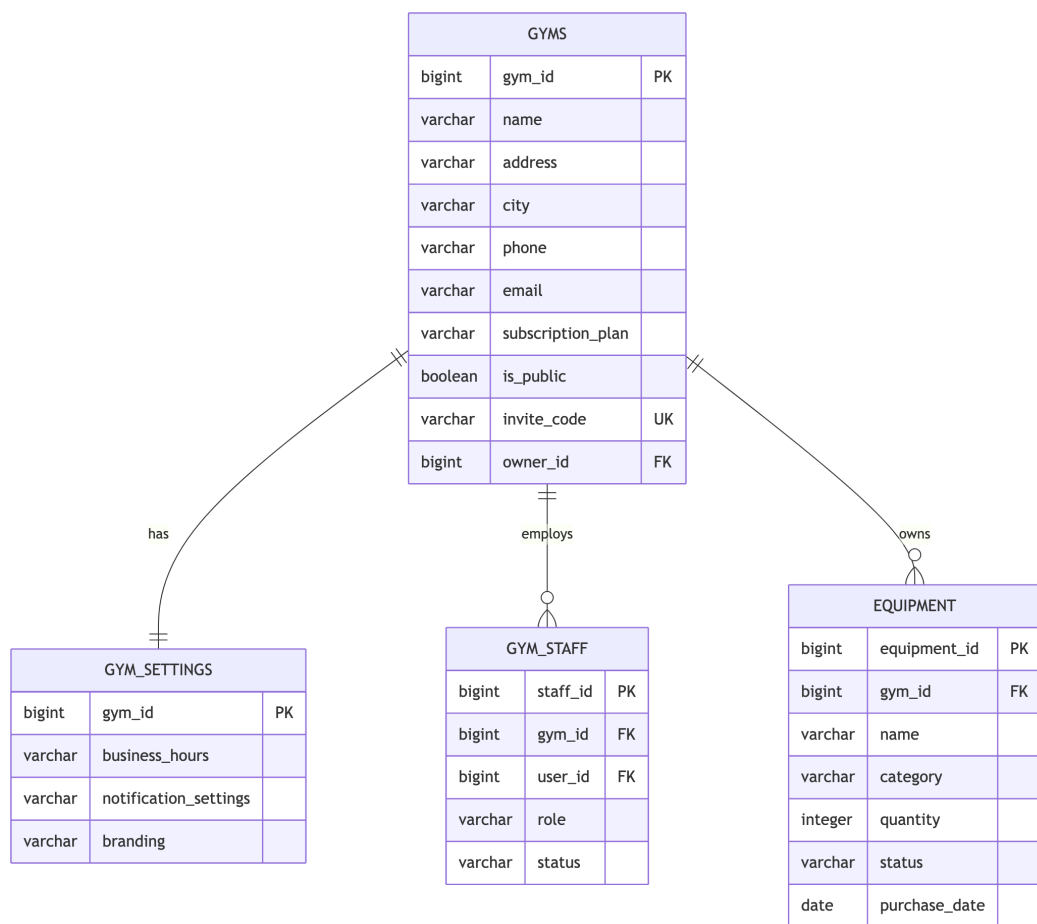
Chapter 5

Entity-Relationship Diagram - Gym Management System

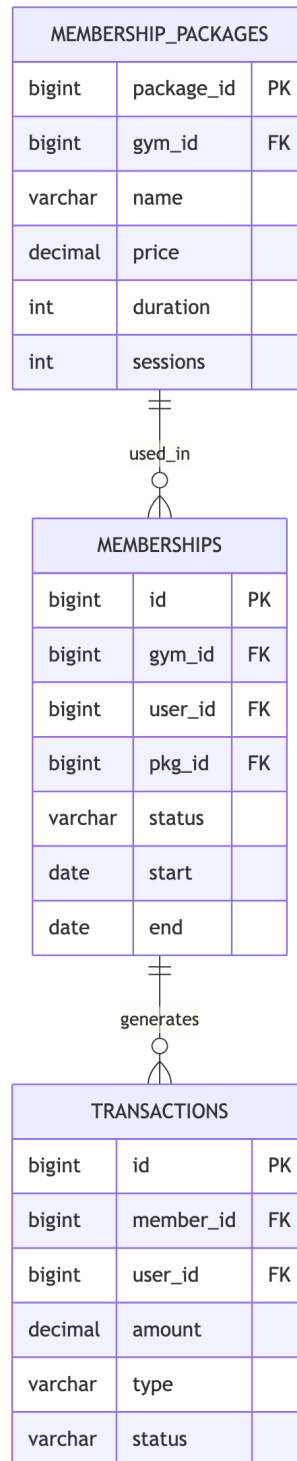
5.1 1. User & Authentication Entities



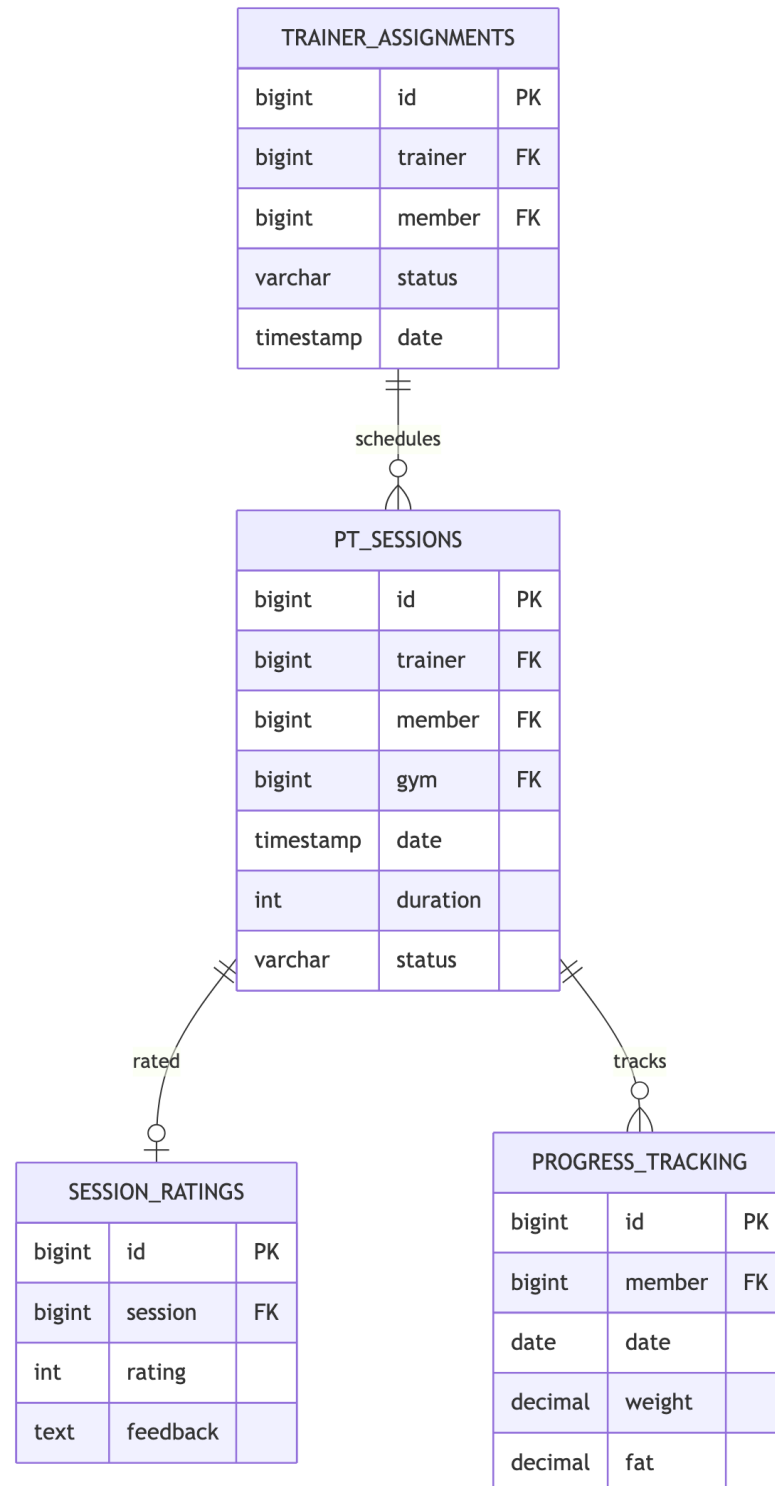
5.2 2. Gym Management Entities



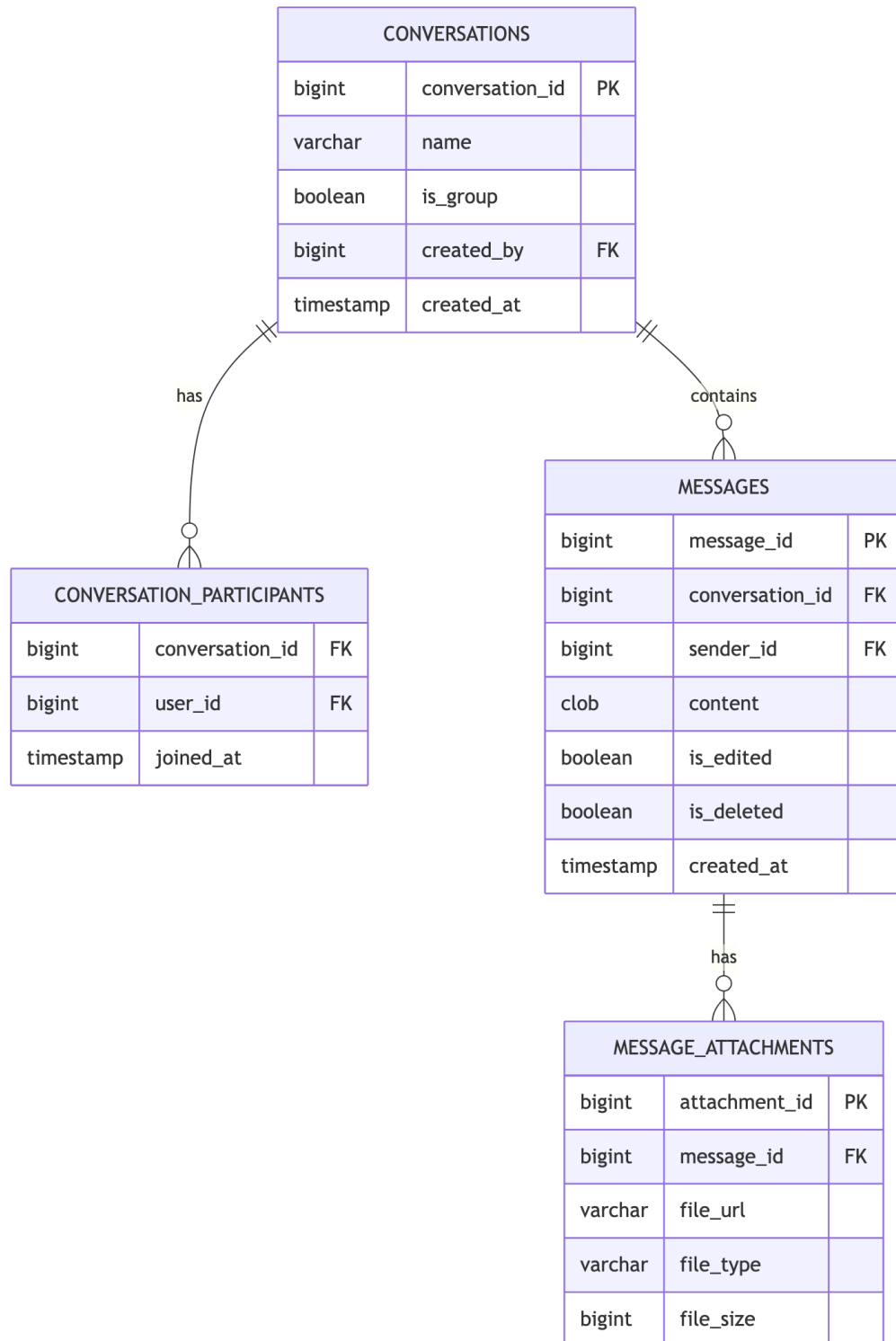
5.3 3. Membership Entities



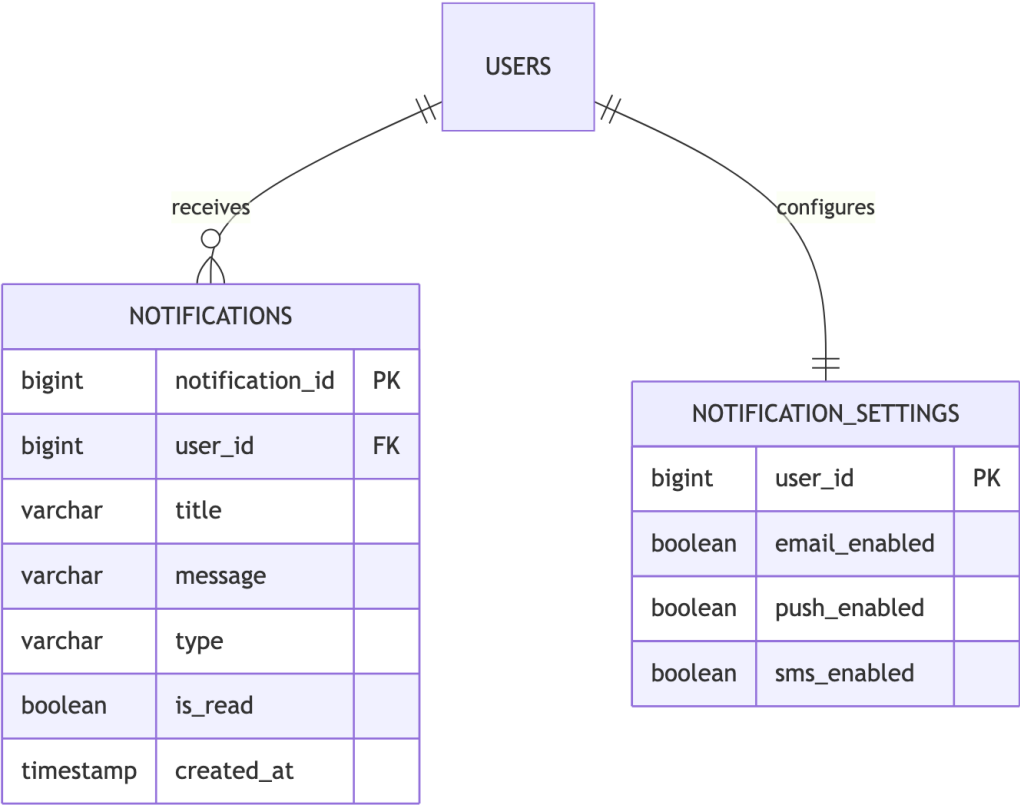
5.4 4. Training & Session Entities



5.5 5. Communication Entities

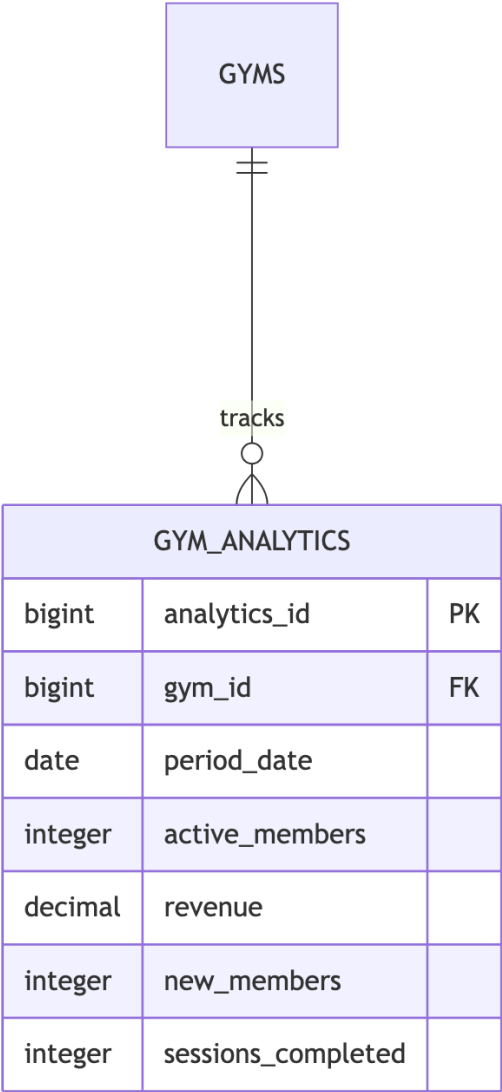


5.6 6. Notification Entities



5.7 7. Analytics Entities

TRAINER_STATS		
bigint	stats_id	PK
bigint	trainer_id	FK
date	period_date	
integer	sessions_count	
decimal	avg_rating	
integer	active_clients	



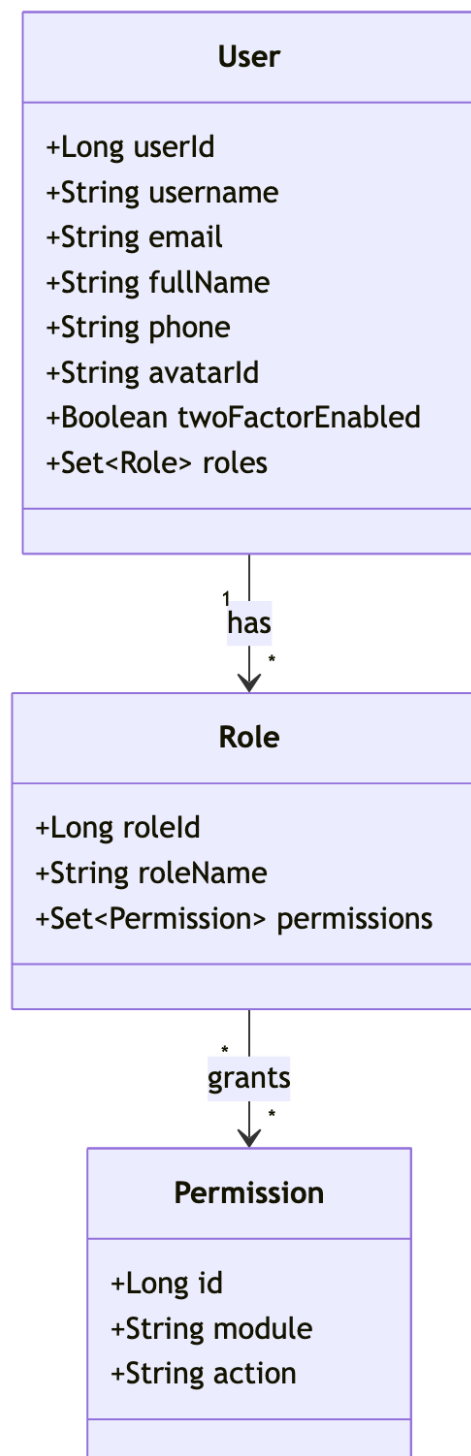
5.8 Entity Summary Table

Category	Entities	Key Relationships
Auth	Users, Roles, Permissions	Many-to-many via mapping tables
Gym	Gyms, Settings, Staff, Equipment	Owner-owned, one-to-many
Members	Packages, Memberships, Transactions	Package → Membership → Transaction
Training	Assignments, Sessions, Ratings, Progress	Trainer-Member assignments
Chat	Conversations, Messages, Attachments	Conversation hierarchy
Notify	Notifications, Settings	User notifications
Analytics	Gym Analytics, Trainer Stats	Period-based aggregations

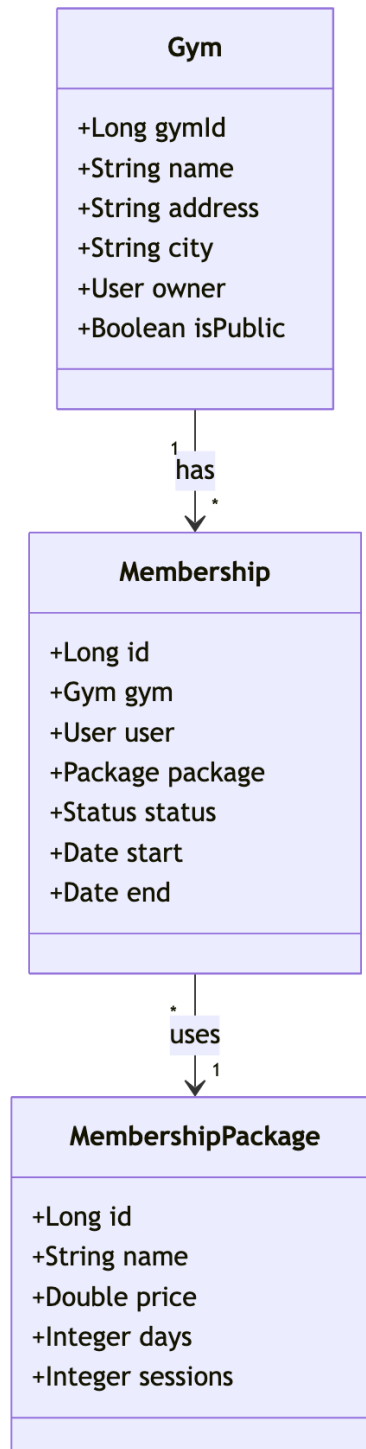
Chapter 6

UML Diagrams - Gym Management System

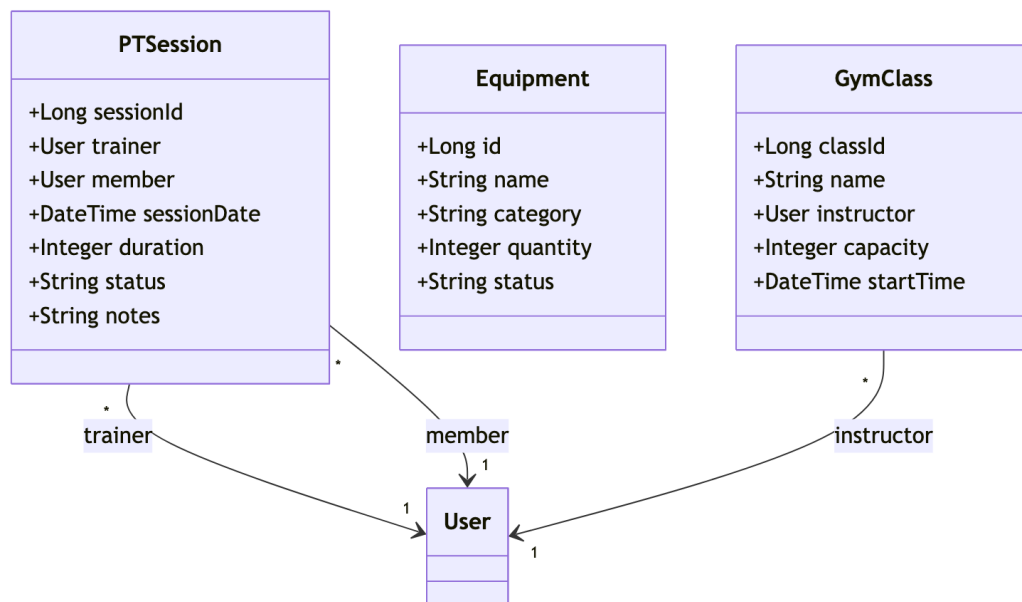
6.1 1. Class Diagram - Core User & Auth



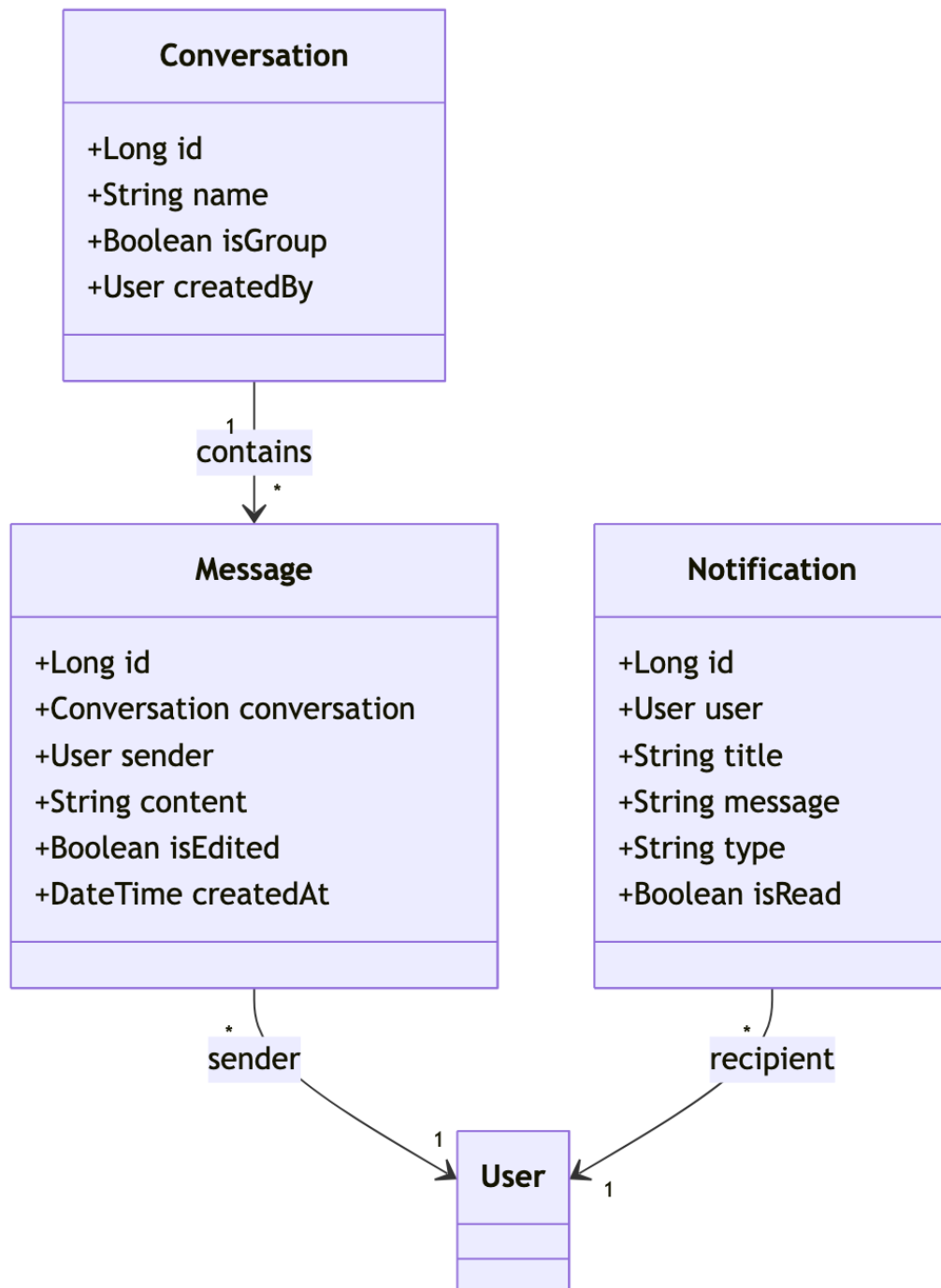
6.2 2. Class Diagram - Gym & Membership



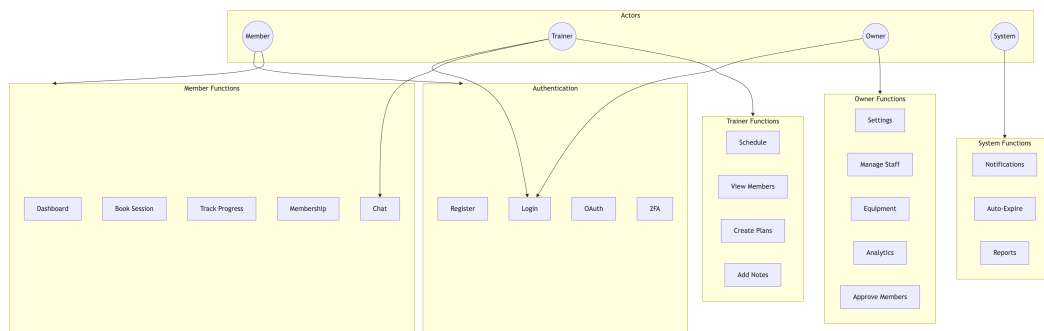
6.3 3. Class Diagram - Training & Sessions



6.4 4. Class Diagram - Communication

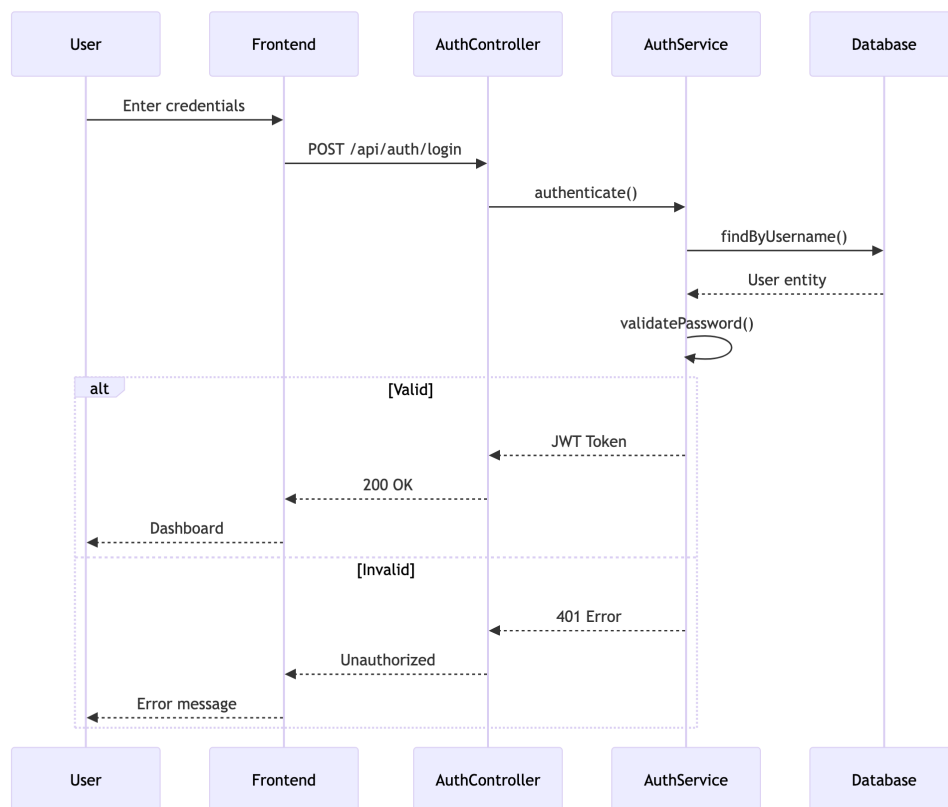


6.5 5. Use Case Diagram

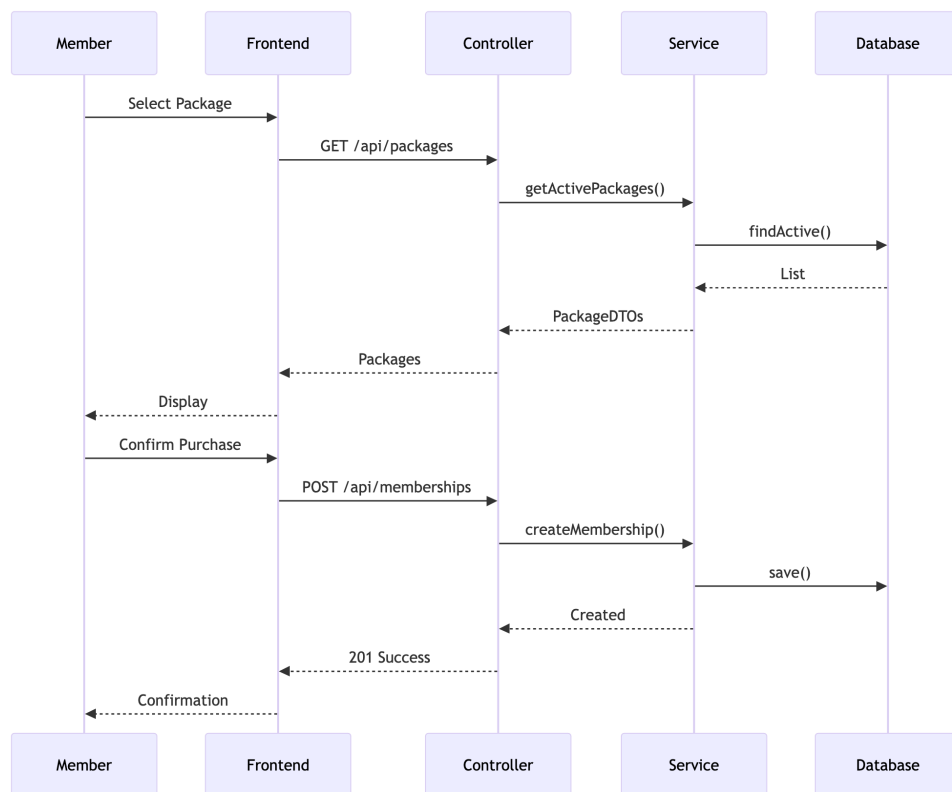


6.6 6. Sequence Diagrams

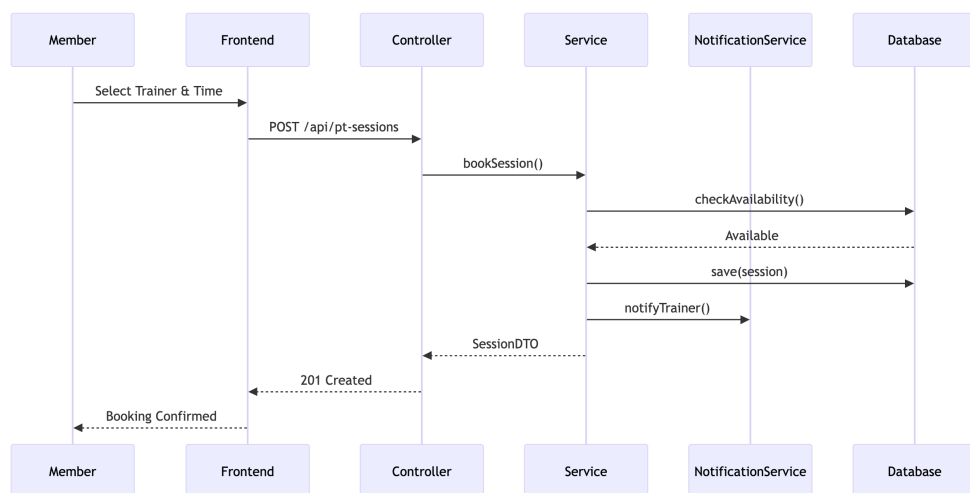
6.6.1 6.1 User Authentication Flow



6.6.2 6.2 Membership Purchase Flow

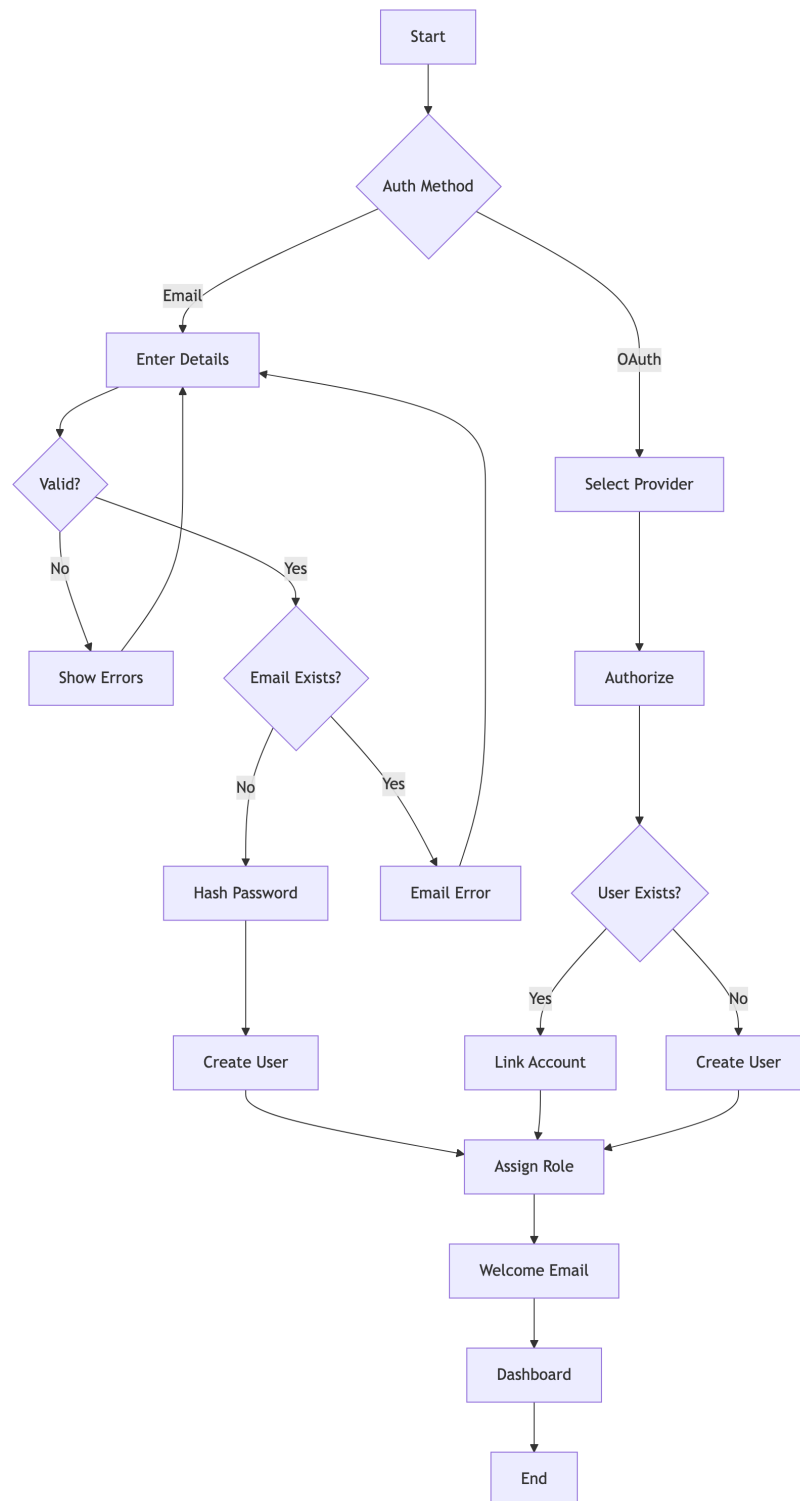


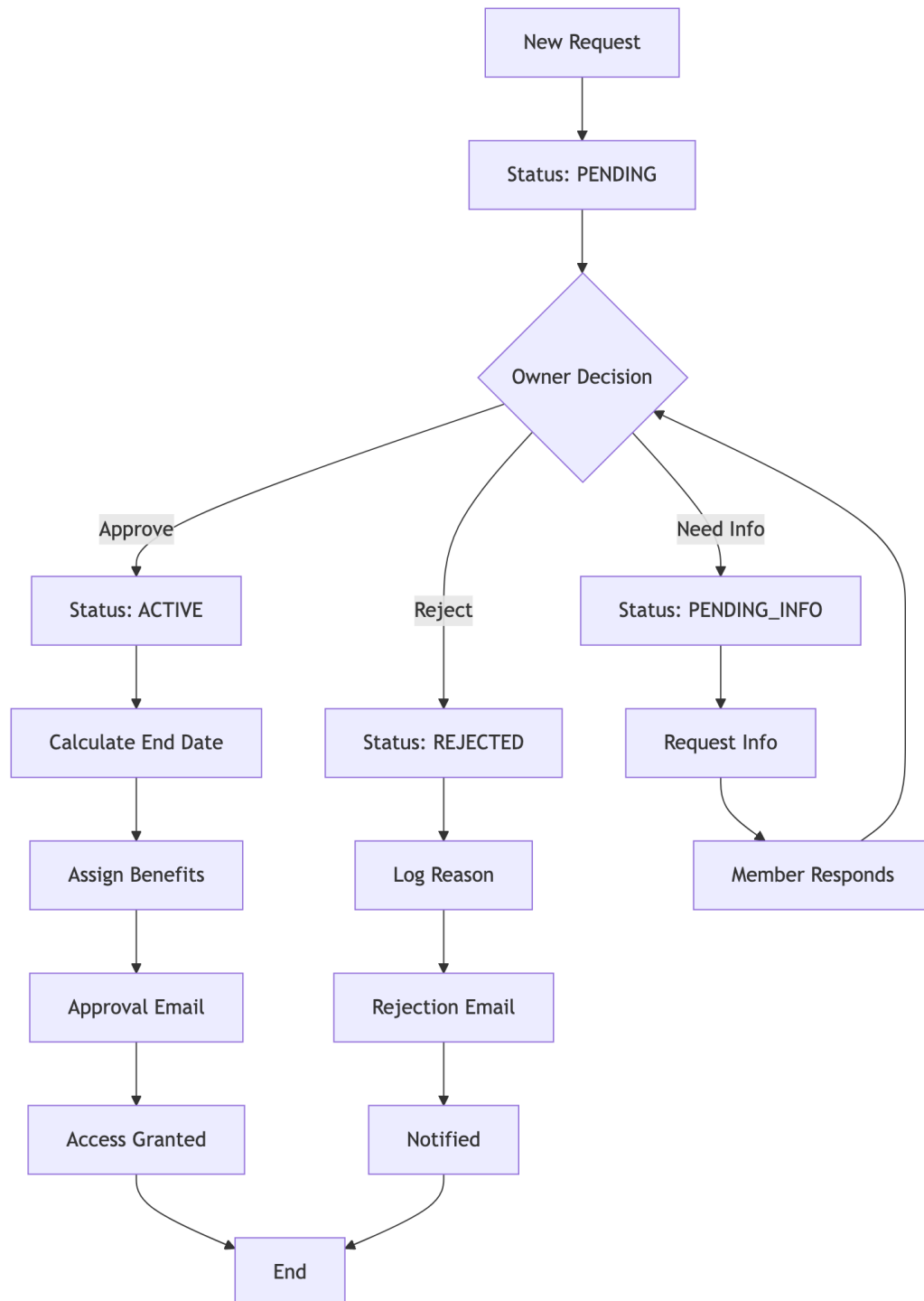
6.6.3 6.3 PT Session Booking Flow

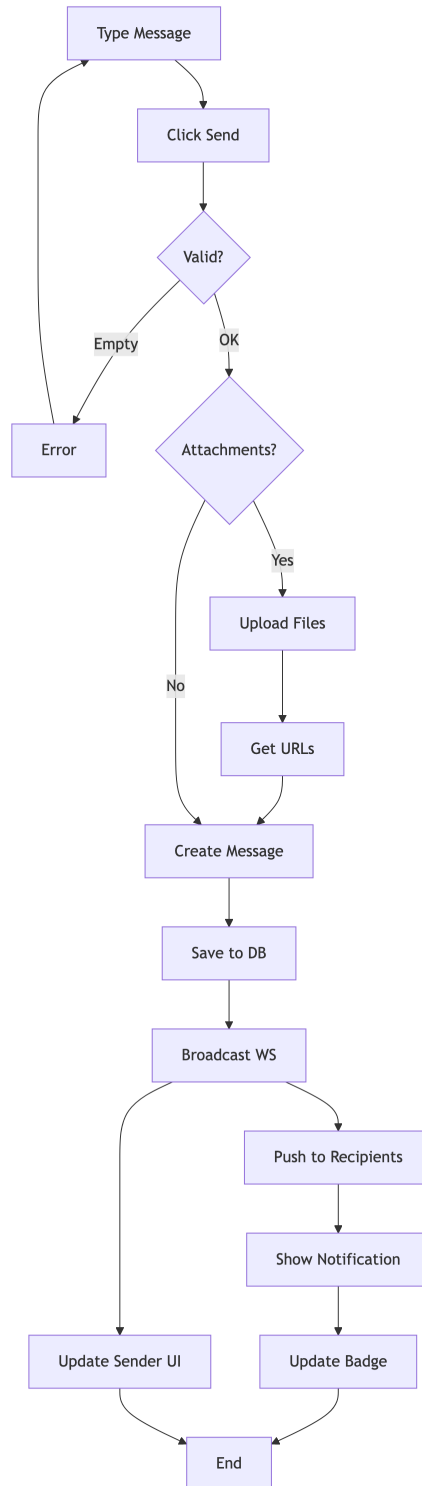


6.7 7. Activity Diagrams

6.7.1 7.1 User Registration Process



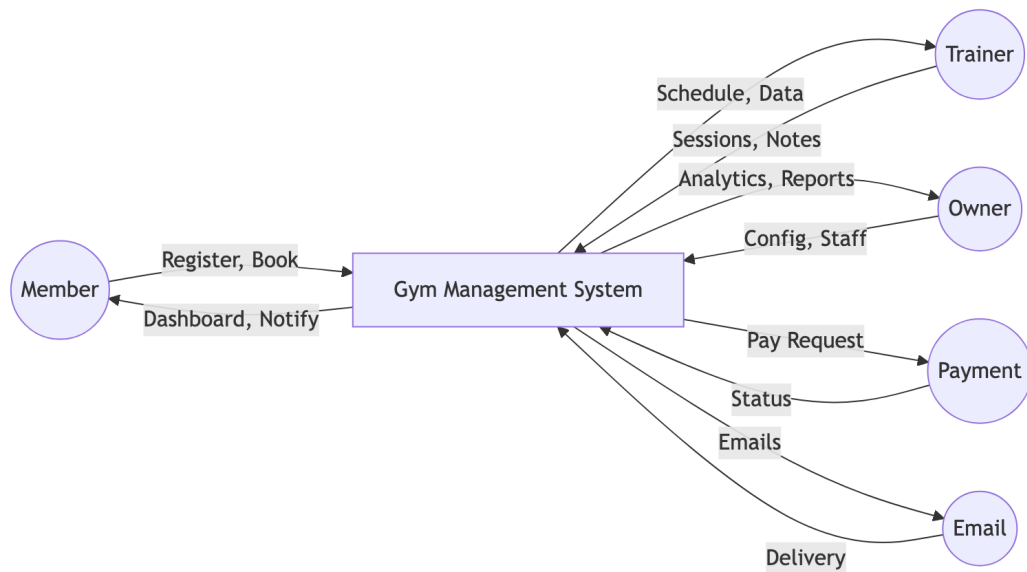
6.7.2 7.2 Membership Approval Workflow

6.7.3 7.3 Chat Message Flow

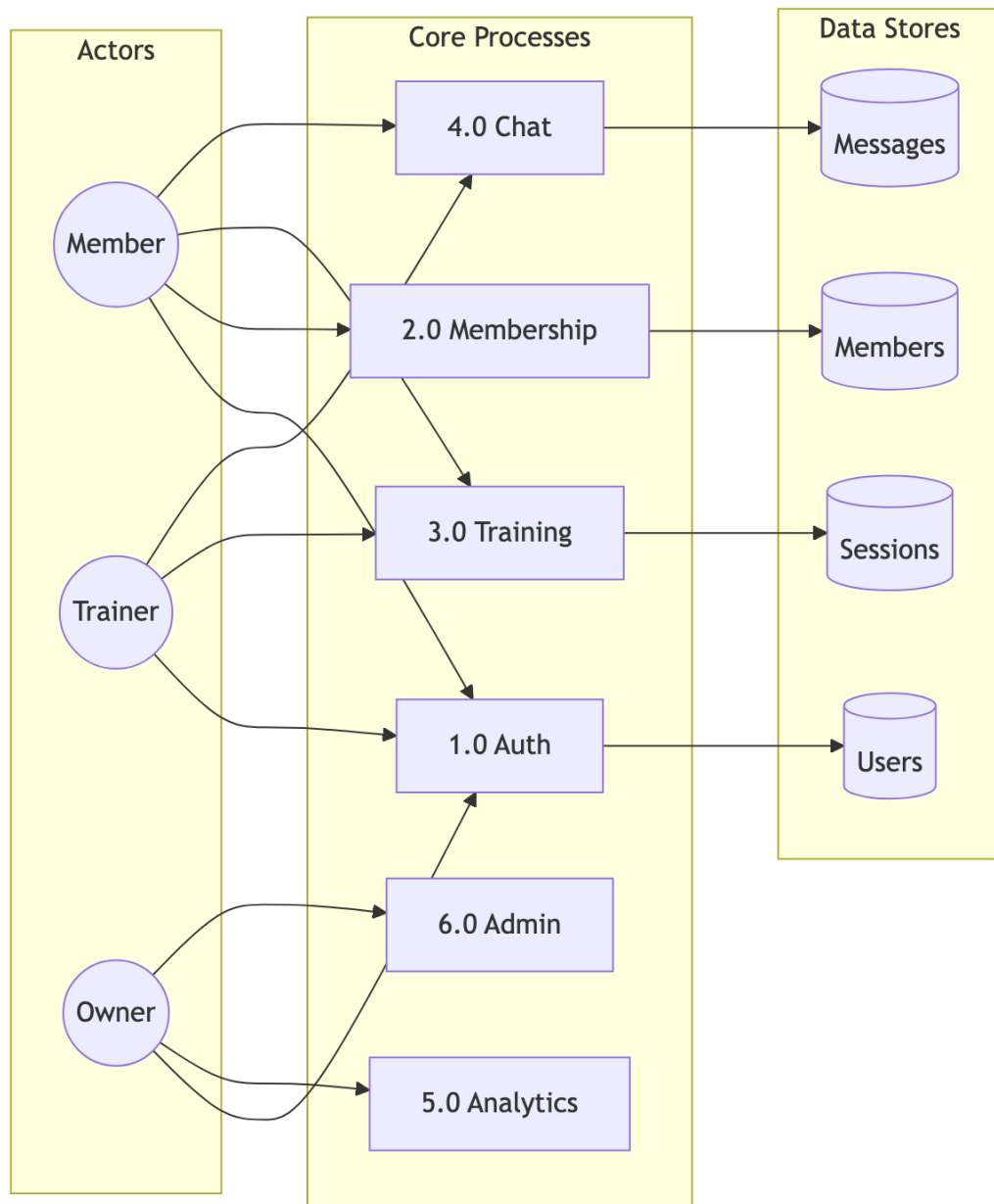
Chapter 7

Data Flow Diagrams - Gym Management System

7.1 Level 0: Context Diagram

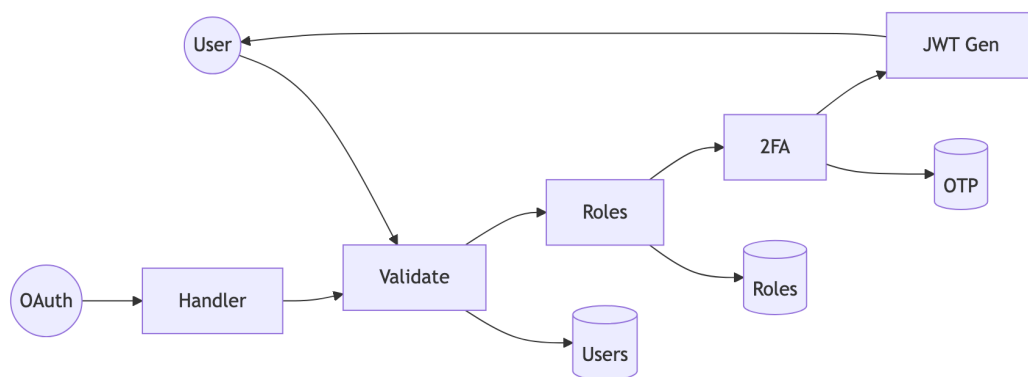


7.2 Level 1: Main Processes

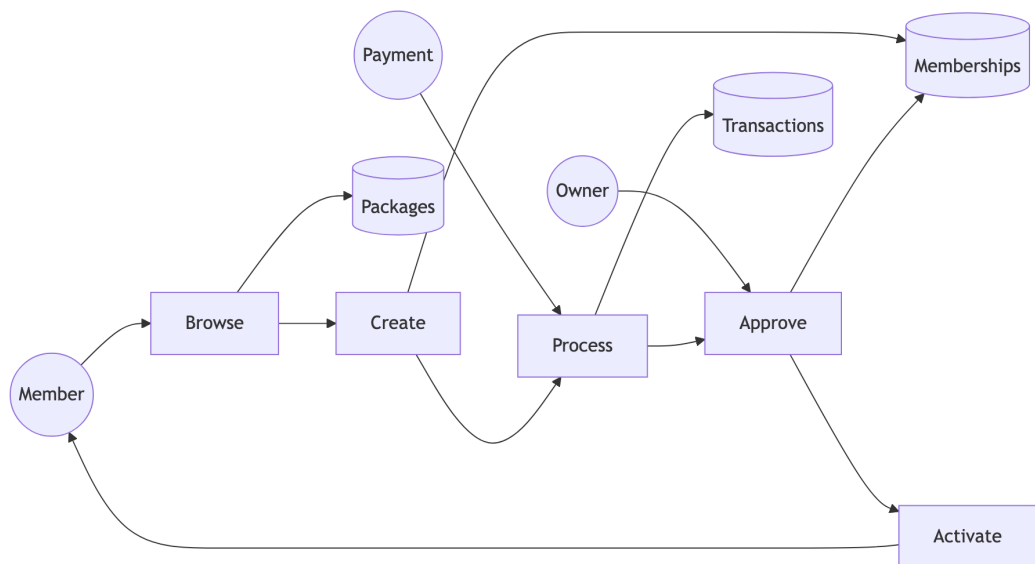


7.3 Level 2: Detailed Sub-Processes

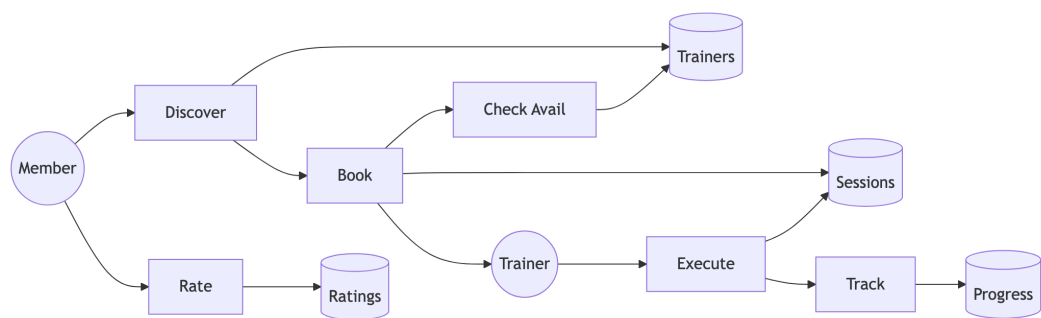
7.3.1 2.1 Authentication (Process 1.0)



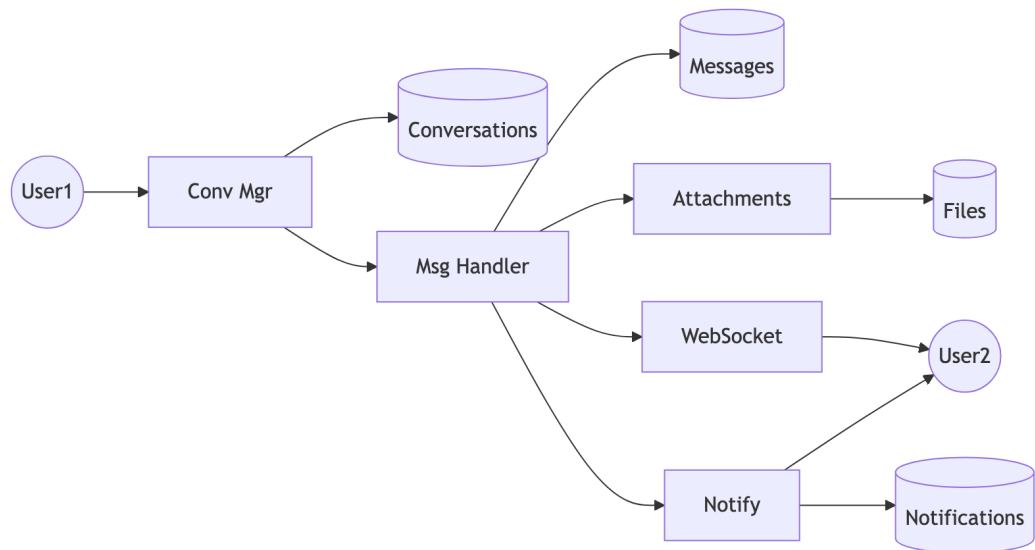
7.3.2 2.2 Membership (Process 2.0)



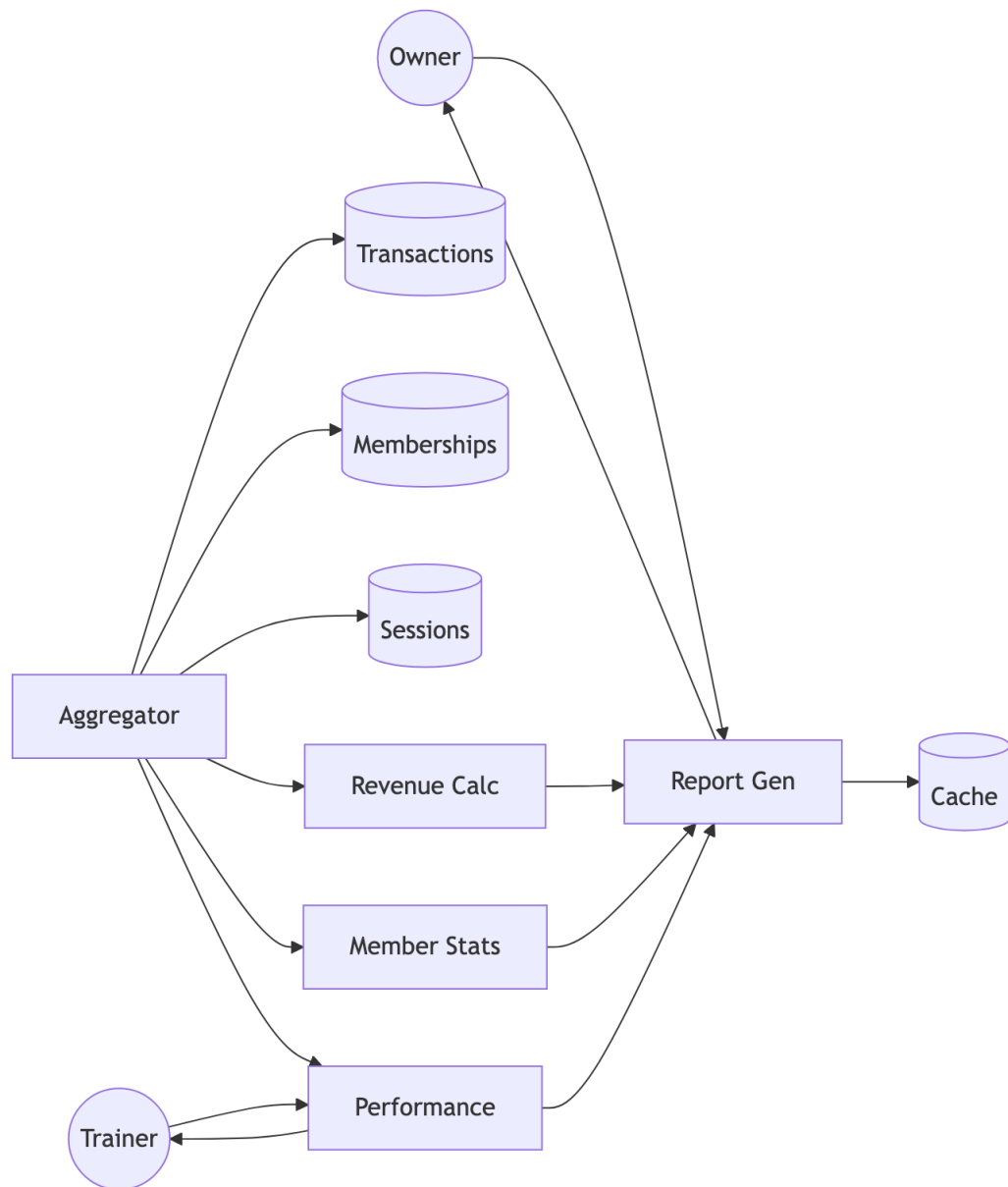
7.3.3 2.3 Training (Process 3.0)



7.3.4 2.4 Communication (Process 4.0)



7.3.5 2.5 Analytics (Process 5.0)



7.4 Data Dictionary

Store	Description	Key Fields
Users	System users	user_id, email, roles
Memberships	Subscriptions	id, user_id, status
Sessions	PT records	id, trainer_id, date
Messages	Chat messages	id, content
Transactions	Payments	id, amount, status

Chapter 8

Algorithms - Gym Management System

8.1 1. JWT Authentication Algorithm

8.1.1 Token Generation

ALGORITHM: JWT_Token_Generation

INPUT: User user, Set<Role> roles

OUTPUT: String jwtToken

BEGIN

 // Create claims payload

 claims = {

 "sub": user.username,

 "userId": user.userId,

 "email": user.email,

 "roles": roles.map(r -> r.roleName),

 "iat": currentTimestamp(),

 "exp": currentTimestamp() + TOKEN_VALIDITY_MS

 }

 // Sign with HMAC-SHA512

 header = {

 "alg": "HS512",

 "typ": "JWT"

 }

 signature = HMAC_SHA512(

 base64Encode(header) + "." + base64Encode(claims),

```
        SECRET_KEY
    )

    RETURN base64Encode(header) + "." +
           base64Encode(claims) + "." +
           base64Encode(signature)

END
```

8.1.2 Token Validation

ALGORITHM: JWT_Token_Validation

INPUT: String token

OUTPUT: Claims claims OR Exception

```
BEGIN

    parts = token.split(".")

    IF parts.length != 3 THEN
        THROW InvalidTokenException
    END IF

    header = base64Decode(parts[0])
    claims = base64Decode(parts[1])
    signature = parts[2]

    // Verify signature
    expectedSig = base64Encode(HMAC_SHA512(
        parts[0] + "." + parts[1],
        SECRET_KEY
    ))

    IF signature != expectedSig THEN
        THROW SignatureVerificationException
    END IF
END
```

```
END IF

// Check expiration
IF claims.exp < currentTimestamp() THEN
    THROW TokenExpiredException
END IF

RETURN claims
END
```

8.2 2. Role-Based Access Control (RBAC) Algorithm

8.2.1 Permission Check

```
ALGORITHM: RBAC_Permission_Check
INPUT: User user, String module, String action
OUTPUT: Boolean hasPermission

BEGIN
    // Get all user roles
    userRoles = user.getRoles()

    // Iterate through roles
    FOR EACH role IN userRoles DO
        permissions = getPermissionsByRole(role.roleId)

        FOR EACH permission IN permissions DO
            IF permission.module == module AND
               permission.action == action THEN
                RETURN TRUE
            END IF
        END FOR
    END FOR

    // Check wildcard permissions
```

```
        IF permission.module == "*" OR
           permission.action == "*" THEN
            IF matchesWildcard(permission, module, action) THEN
                RETURN TRUE
            END IF
        END IF
    END FOR
END FOR

RETURN FALSE
END

FUNCTION matchesWildcard(permission, module, action):
    IF permission.module == "*" THEN
        RETURN permission.action == action OR
               permission.action == "*"
    END IF
    IF permission.action == "*" THEN
        RETURN permission.module == module
    END IF
    RETURN FALSE
END FUNCTION
```

8.2.2 Role Hierarchy

```
ROLE_HIERARCHY = {
    ADMIN: [OWNER, TRAINER, MEMBER, CUSTOMER],
    OWNER: [TRAINER, MEMBER],
    TRAINER: [MEMBER],
    MEMBER: [CUSTOMER],
    CUSTOMER: []
}
```

ALGORITHM: Check_Role_Hierarchy

INPUT: User user, String requiredRole

OUTPUT: Boolean hasRole

BEGIN

 userRoles = user.getRoles()

 FOR EACH role IN userRoles DO

 IF role.name == requiredRole THEN

 RETURN TRUE

 END IF

 // Check inherited roles

 inheritedRoles = ROLE_HIERARCHY[role.name]

 IF requiredRole IN inheritedRoles THEN

 RETURN TRUE

 END IF

 END FOR

 RETURN FALSE

END

8.3 3. Membership Package Assignment Algorithm

ALGORITHM: Assign_Membership

INPUT: User member, MembershipPackage package, Gym gym

OUTPUT: Membership membership

BEGIN

 // Validate inputs

 IF member == NULL OR package == NULL OR gym == NULL THEN

 THROW InvalidInputException

 END IF

```
// Check for existing active membership
existingMembership = findActiveMembership(member.userId, gym.gymId)

IF existingMembership != NULL THEN
    IF existingMembership.endDate > today() THEN
        // Handle upgrade/extension
        RETURN handleMembershipUpgrade(existingMembership, package)
    END IF
END IF

// Calculate dates
startDate = today()
endDate = startDate.plusDays(package.durationDays)

// Create new membership
membership = new Membership()
membership.user = member
membership.gym = gym
membership.membershipPackage = package
membership.startDate = startDate
membership.endDate = endDate
membership.status = PENDING
membership.createdAt = now()

// Assign PT session credits
IF package.includedPTSessions > 0 THEN
    createPTSessionCredits(member, package.includedPTSessions)
END IF

// Save and return
save(membership)
```

```
        sendNotification(gym.owner, "New membership request")

    RETURN membership

END

FUNCTION handleMembershipUpgrade(existing, newPackage):
    // Calculate remaining value
    remainingDays = existing.endDate - today()
    remainingValue = (remainingDays / existing.package.durationDays)
                    * existing.package.price

    // Apply credit to new package
    newPrice = newPackage.price - remainingValue

    // Extend end date
    existing.membershipPackage = newPackage
    existing.endDate = today().plusDays(newPackage.durationDays)

    save(existing)
    RETURN existing
END FUNCTION
```

8.4 4. Session Rating Calculation Algorithm

```
ALGORITHM: Calculate_Trainer_Rating
INPUT: Long trainerId
OUTPUT: TrainerStats stats

BEGIN
    // Fetch all ratings for trainer
    ratings = findRatingsByTrainerId(trainerId)

    IF ratings.isEmpty() THEN
```



```
        RETURN TrainerStats(  
            averageRating: 0.0,  
            totalReviews: 0,  
            ratingDistribution: {}  
        )  
    END IF  
  
    // Calculate statistics  
    totalSum = 0  
    distribution = {1: 0, 2: 0, 3: 0, 4: 0, 5: 0}  
  
    FOR EACH rating IN ratings DO  
        totalSum = totalSum + rating.score  
        distribution[rating.score]++  
    END FOR  
  
    averageRating = totalSum / ratings.size()  
  
    // Round to 1 decimal place  
    averageRating = ROUND(averageRating * 10) / 10  
  
    // Calculate weighted score (recent ratings worth more)  
    weightedSum = 0  
    weightTotal = 0  
  
    FOR i = 0 TO ratings.size() - 1 DO  
        weight = 1 + (i * 0.1) // Recent ratings weighted higher  
        weightedSum = weightedSum + (ratings[i].score * weight)  
        weightTotal = weightTotal + weight  
    END FOR  
  
    weightedAverage = weightedSum / weightTotal
```

```
    RETURN TrainerStats(  
        averageRating: averageRating,  
        weightedRating: ROUND(weightedAverage * 10) / 10,  
        totalReviews: ratings.size(),  
        ratingDistribution: distribution  
    )  
END
```

8.5 5. Analytics Computation Algorithm

8.5.1 Revenue Calculation

ALGORITHM: Calculate_Revenue_Analytics

INPUT: Long gymId, DateRange period

OUTPUT: RevenueAnalytics analytics

```
BEGIN  
  
    transactions = findTransactionsByGymAndPeriod(gymId, period)  
  
    // Calculate totals  
    totalRevenue = 0  
    revenueByType = {}  
    revenueByDay = {}  
  
    FOR EACH txn IN transactions DO  
        IF txn.status == "COMPLETED" THEN  
            totalRevenue = totalRevenue + txn.amount  
  
            // Group by type  
            type = txn.type  
            revenueByType[type] = (revenueByType[type] OR 0) + txn.amount
```

```
        // Group by day
        day = txn.createdAt.toDate()
        revenueByDay[day] = (revenueByDay[day] OR 0) + txn.amount
    END IF
END FOR

// Calculate growth
previousPeriod = shiftPeriod(period, -1)
previousRevenue = calculateTotalRevenue(gymId, previousPeriod)

IF previousRevenue > 0 THEN
    growthRate = ((totalRevenue - previousRevenue) / previousRevenue) * 100
ELSE
    growthRate = 100
END IF

// Project monthly
daysInPeriod = period.end - period.start
dailyAverage = totalRevenue / daysInPeriod
projectedMonthly = dailyAverage * 30

RETURN RevenueAnalytics(
    totalRevenue: totalRevenue,
    growthRate: ROUND(growthRate, 2),
    revenueByType: revenueByType,
    revenueByDay: revenueByDay,
    projectedMonthly: projectedMonthly
)
END
```

8.5.2 Member Statistics

ALGORITHM: Calculate_Member_Statistics

INPUT: Long gymId

OUTPUT: MemberStats stats

BEGIN

```
memberships = findMembershipsByGym(gymId)
```

```
// Count by status
```

```
activeCount = 0
```

```
pendingCount = 0
```

```
expiredCount = 0
```

```
// Track trends
```

```
newThisMonth = 0
```

```
newLastMonth = 0
```

```
currentMonth = getMonthStart(today())
```

```
lastMonth = getMonthStart(today().minusMonths(1))
```

```
FOR EACH membership IN memberships DO
```

```
    SWITCH membership.status:
```

```
        CASE ACTIVE:
```

```
            activeCount++
```

```
            IF membership.createdAt >= currentMonth THEN
```

```
                newThisMonth++
```

```
            ELSE IF membership.createdAt >= lastMonth THEN
```

```
                newLastMonth++
```

```
            END IF
```

```
        CASE PENDING:
```

```
            pendingCount++
```

```
        CASE EXPIRED:
```

```
            expiredCount++
```

```
    END SWITCH
```

```
END FOR

// Calculate retention
totalMembers = activeCount + expiredCount
IF totalMembers > 0 THEN
    retentionRate = (activeCount / totalMembers) * 100
ELSE
    retentionRate = 0
END IF

// Calculate growth
IF newLastMonth > 0 THEN
    growthRate = ((newThisMonth - newLastMonth) / newLastMonth) * 100
ELSE
    growthRate = newThisMonth > 0 ? 100 : 0
END IF

RETURN MemberStats(
    totalActive: activeCount,
    totalPending: pendingCount,
    totalExpired: expiredCount,
    newThisMonth: newThisMonth,
    retentionRate: ROUND(retentionRate, 2),
    monthlyGrowth: ROUND(growthRate, 2)
)
END
```

8.6 6. Duplicate Package Cleanup Algorithm

ALGORITHM: Cleanup_Duplicate_Packages

INPUT: None (scheduled job)

OUTPUT: Integer deletedCount

```
BEGIN

    allPackages = findAllPackages()

    IF allPackages.size() < 2 THEN
        RETURN 0
    END IF

    // Group by name + duration (case-insensitive)
    groupedPackages = {}

    FOR EACH pkg IN allPackages DO
        key = toLowerCase(trim(pkg.packageName)) + "|" + pkg.durationDays

        IF groupedPackages[key] == NULL THEN
            groupedPackages[key] = []
        END IF

        groupedPackages[key].add(pkg)
    END FOR

    // Find and delete duplicates
    duplicatesToDelete = []

    FOR EACH key, packages IN groupedPackages DO
        IF packages.size() > 1 THEN
            // Sort by ID (keep oldest)
            sortById(packages)

            // Check each duplicate for usage
            FOR i = 1 TO packages.size() - 1 DO
                duplicate = packages[i]

                memberCount = countMembersByPackage(duplicate.packageId)
```

```
        IF memberCount == 0 THEN
            duplicatesToDelete.add(duplicate.packageId)
        END IF
    END FOR

    END IF

    END FOR

    // Delete unused duplicates
    IF duplicatesToDelete.size() > 0 THEN
        deletePackagesByIds(duplicatesToDelete)
    END IF

    RETURN duplicatesToDelete.size()

END
```

8.7 7. Password Hashing Algorithm

ALGORITHM: Hash_Password

INPUT: String plainPassword

OUTPUT: String hashedPassword

```
BEGIN

    // BCrypt configuration
    SALT_ROUNDS = 12

    // Generate salt
    salt = generateRandomBytes(16)
    salt = bcryptSalt(SALT_ROUNDS)

    // Hash password
    hash = bcrypt(plainPassword, salt)
```

```
// Output format: $2a$12$<22-char-salt><31-char-hash>
RETURN hash
END

ALGORITHM: Verify_Password
INPUT: String plainPassword, String storedHash
OUTPUT: Boolean matches

BEGIN
    // BCrypt handles salt extraction internally
    // Salt is embedded in first 29 characters

    computedHash = bcrypt(plainPassword,
                          extractSalt(storedHash))

    // Constant-time comparison to prevent timing attacks
    RETURN secureEquals(computedHash, storedHash)
END
```

8.8 Algorithm Complexity Analysis

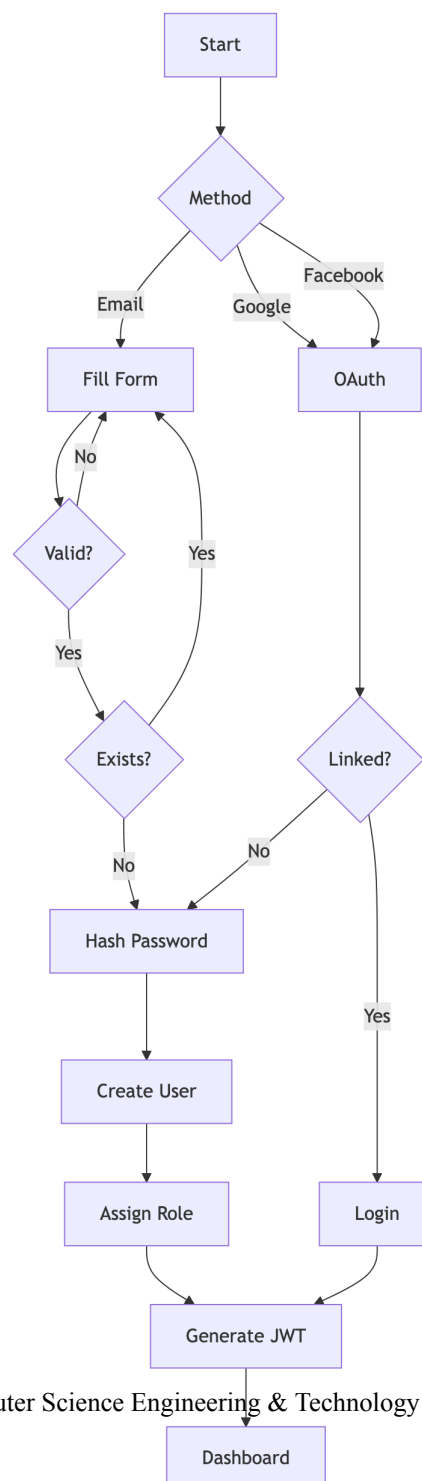
Algorithm	Time Complexity	Space Complexity
JWT Generation	$O(1)$	$O(n)$ where n = claims size
JWT Validation	$O(1)$	$O(1)$
RBAC Check	$O(r \times p)$	$O(1)$ where r = roles, p = permissions
Membership Assignment	$O(1)$	$O(1)$
Rating Calculation	$O(n)$	$O(1)$ where n = ratings count
Revenue Analytics	$O(t)$	$O(d)$ where t = transactions, d = days
Duplicate Cleanup	$O(n \log n)$	$O(n)$ where n = packages
BCrypt Hash	$O(2^{\text{rounds}})$	$O(1)$

Chapter 9

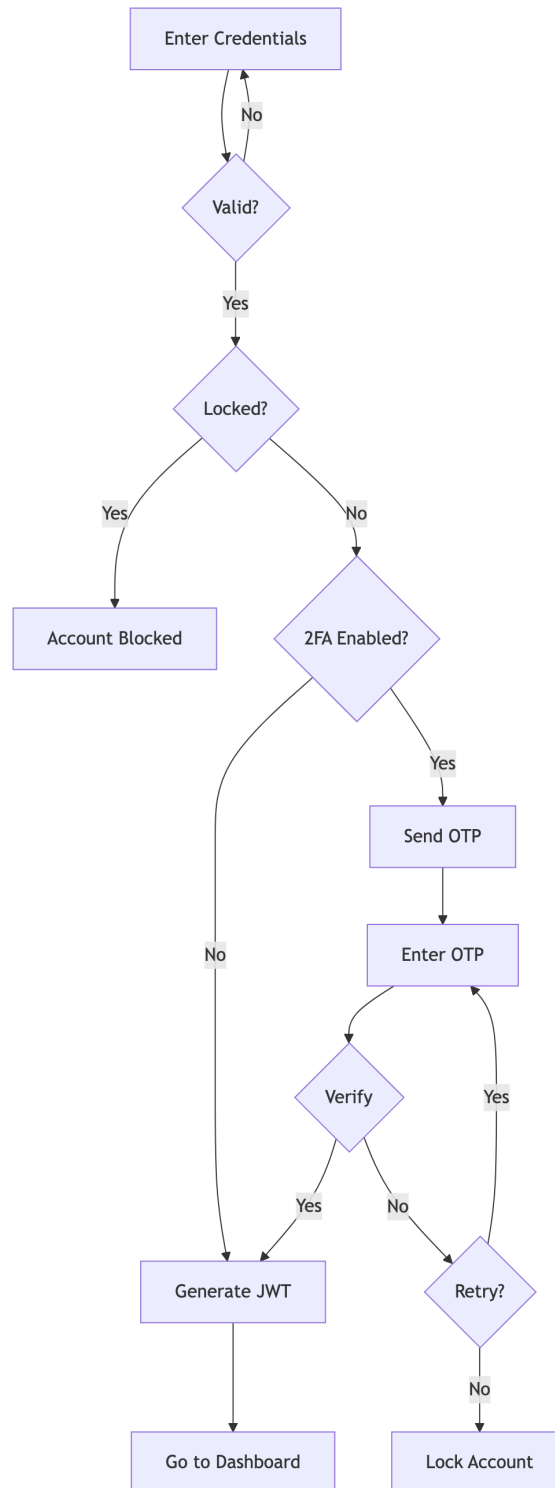
System Workflows - Gym Management System

9.1 1. User Registration & Authentication

9.1.1 1.1 Registration Flow

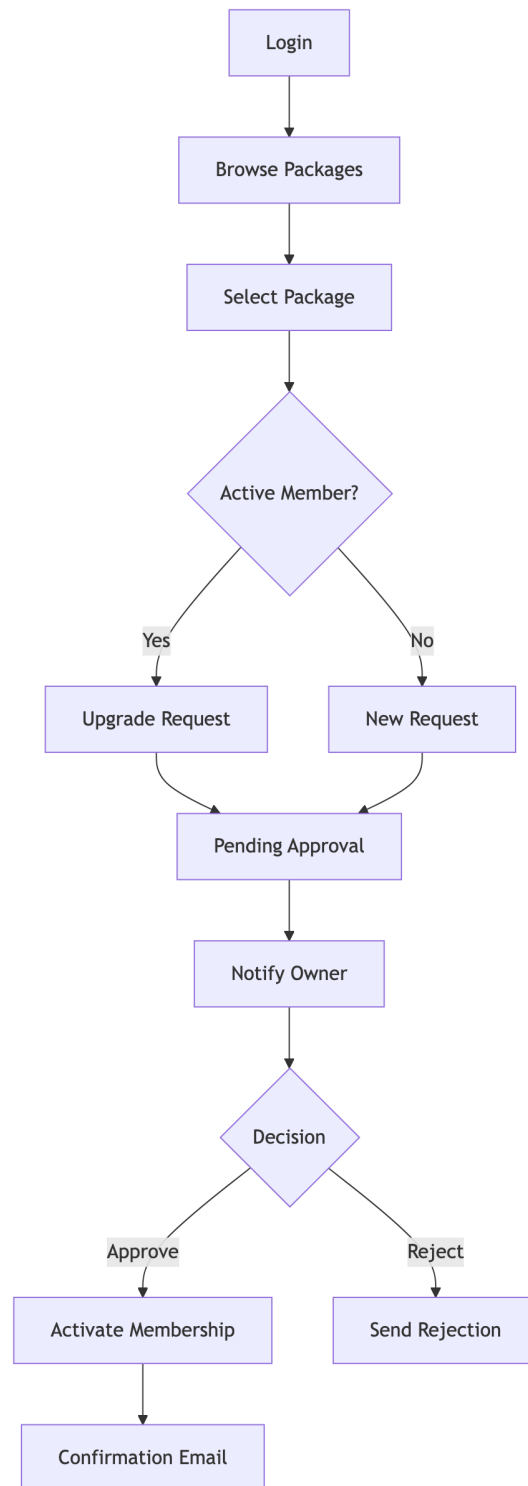


9.1.2 1.2 Login with 2FA

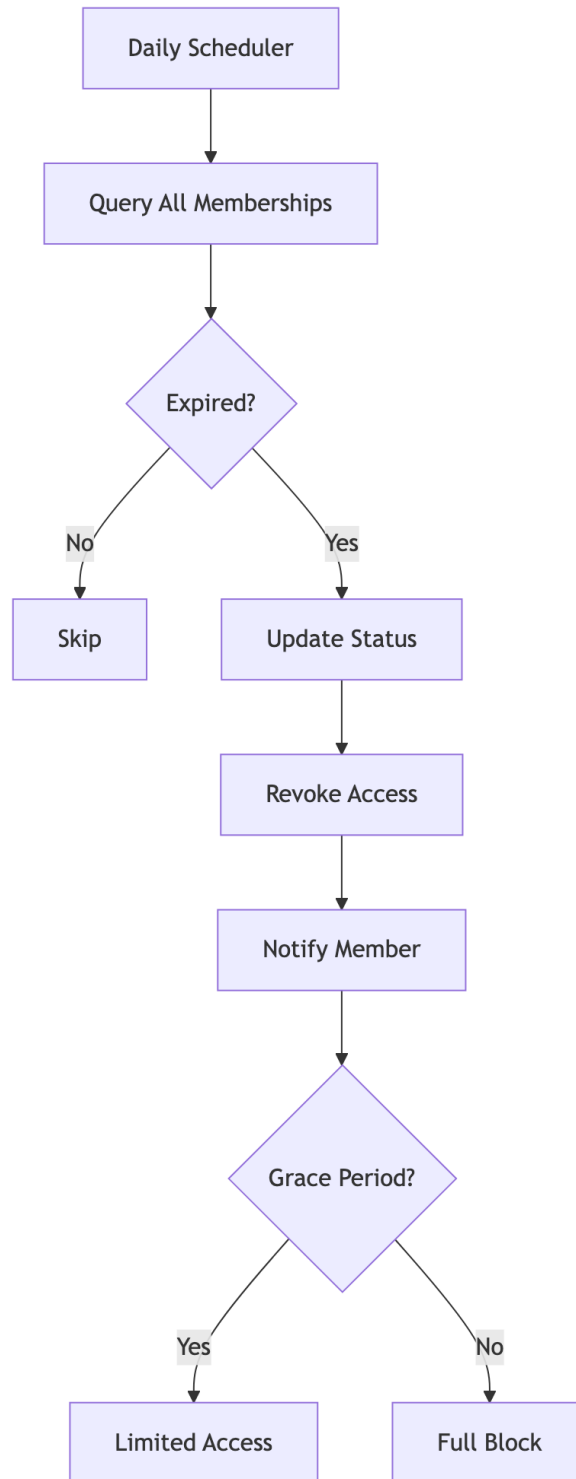


9.2 2. Membership Management

9.2.1 2.1 Purchase Flow

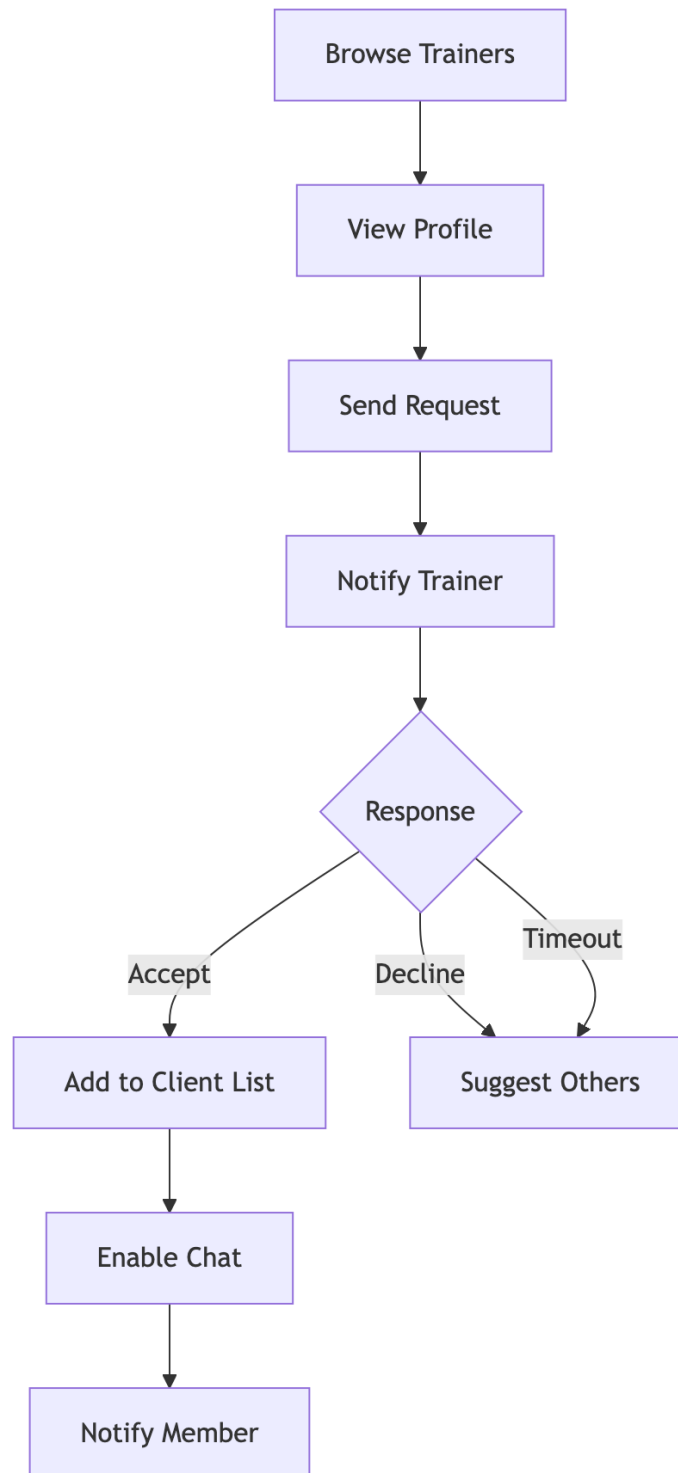


9.2.2 2.2 Expiry Handling

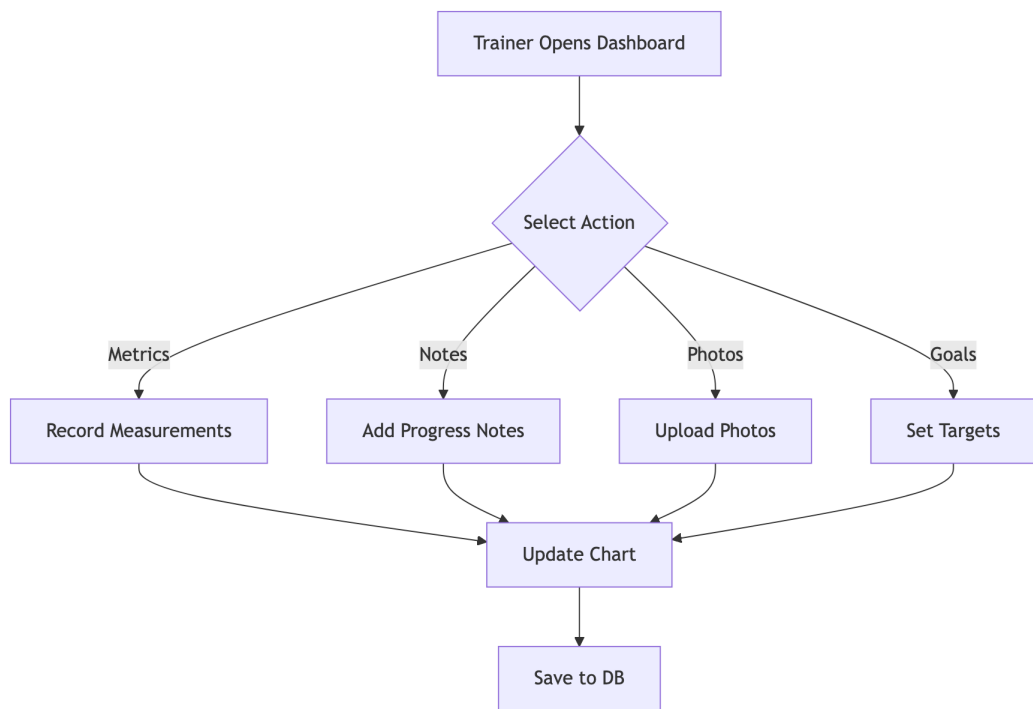


9.3 3. Trainer-Member Interaction

9.3.1 3.1 Trainer Assignment

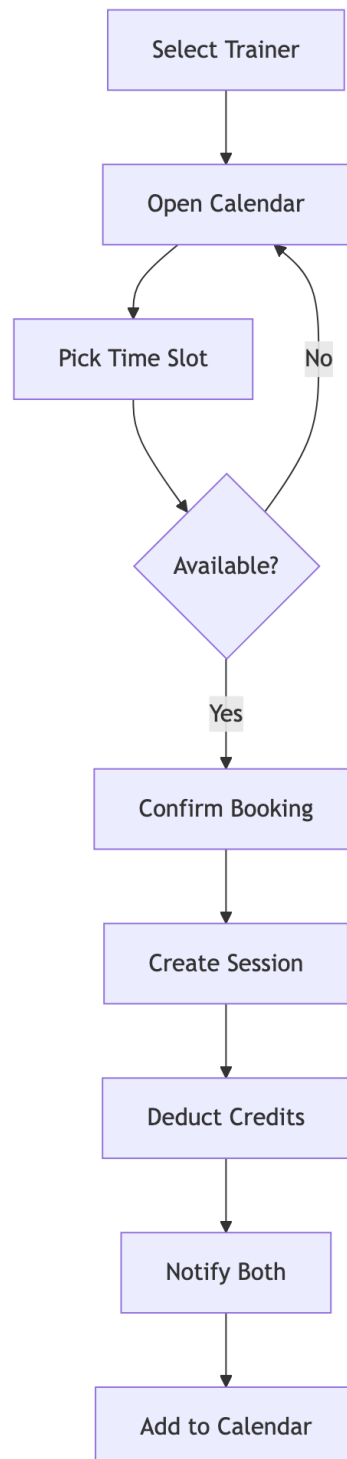


9.3.2 3.2 Progress Tracking

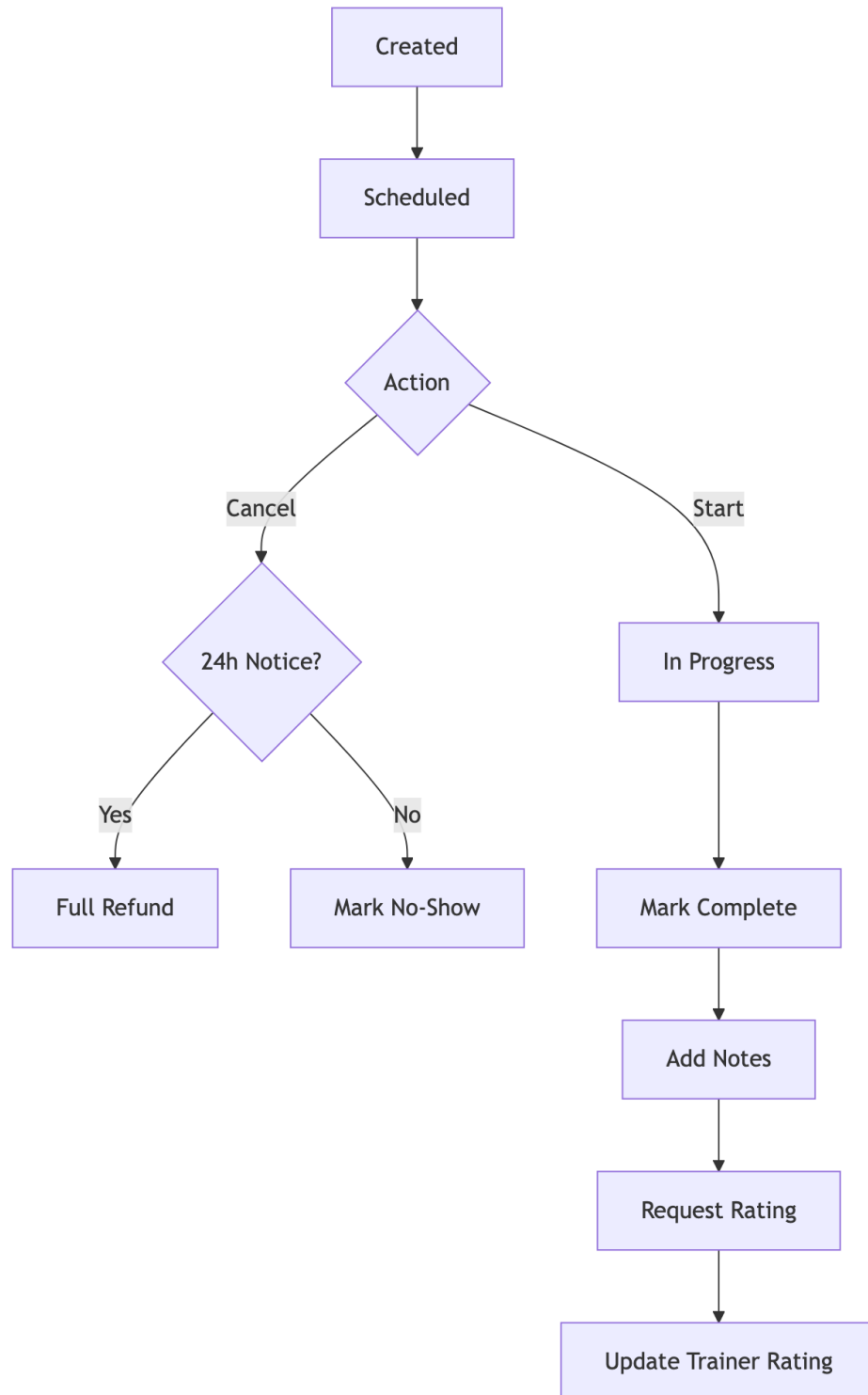


9.4 4. PT Session Booking

9.4.1 4.1 Session Creation

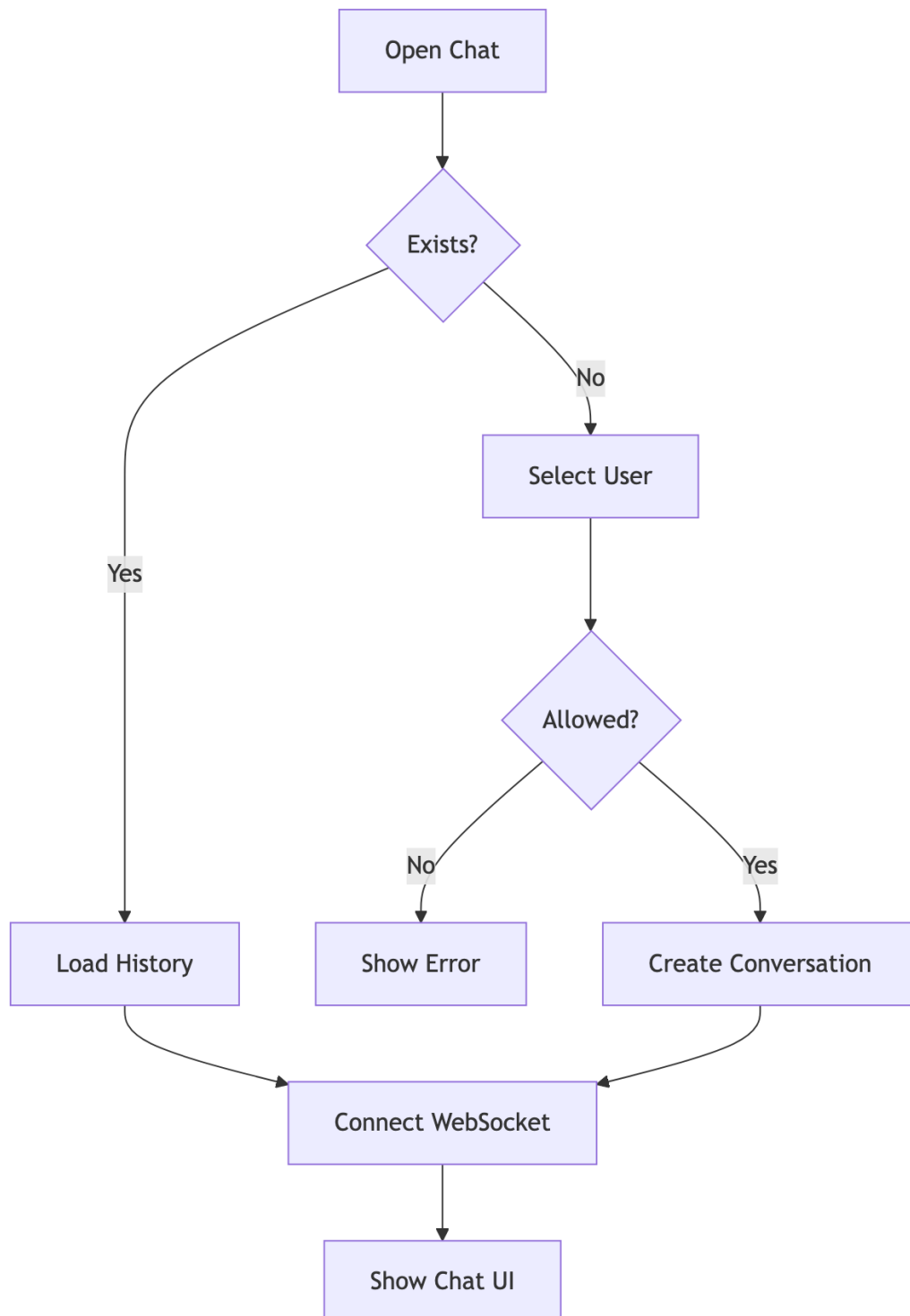


9.4.2 4.2 Session Lifecycle

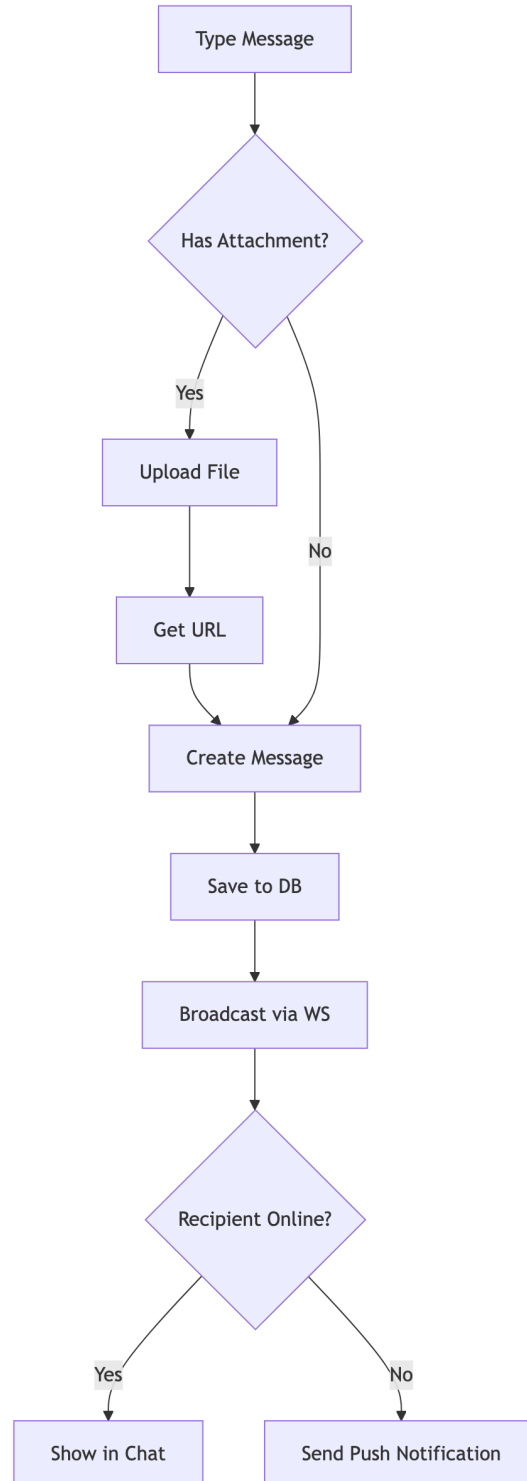


9.5 5. Chat/Messaging

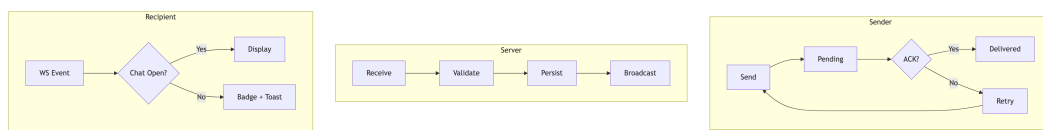
9.5.1 5.1 Conversation



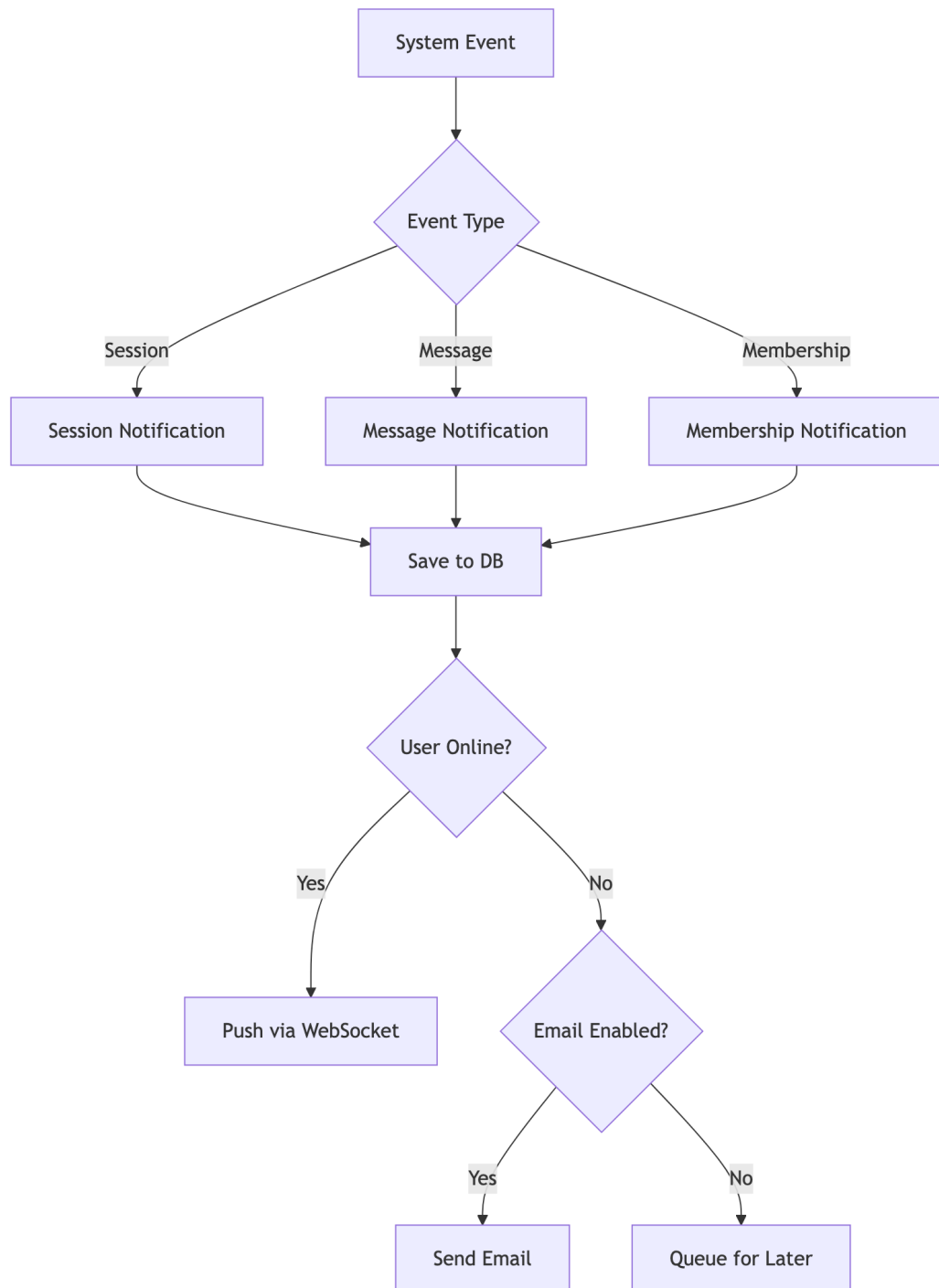
9.5.2 5.2 Message Flow



9.5.3 5.3 Message Delivery Status



9.6 6. Notifications



Chapter 10

Features & Functionality - Gym Management System

10.1 Functional Requirements

10.1.1 1. User Management

ID	Feature	Description	Priority
F1.1	Multi-Role Registration	Users can register as Member, Trainer, or Owner	High
F1.2	OAuth Integration	Login via Google, Facebook	High
F1.3	2FA Support	Email/SMS OTP verification	Medium
F1.4	Profile Management	Edit personal info, avatar, preferences	High
F1.5	Password Reset	Secure password recovery flow	High
F1.6	Account Status	Active, Suspended, Deleted states	Medium

10.1.2 2. Gym Administration

ID	Feature	Description	Priority
F2.1	Gym Creation	Owners can create and configure gyms	High
F2.2	Staff Management	Add/remove trainers, set roles	High
F2.3	Equipment Tracking	Inventory, maintenance scheduling	Medium

ID	Feature	Description	Priority
F2.4	Class Management	Group fitness scheduling	Medium
F2.5	Settings & Branding	Customize gym appearance	Low
F2.6	Invite System	Private gym invite codes	Medium

10.1.3 3. Membership Management

ID	Feature	Description	Priority
F3.1	Package Creation	Define membership packages with pricing	High
F3.2	Membership Purchase	Members can buy/renew memberships	High
F3.3	Approval Workflow	Owner approves membership requests	High
F3.4	Expiry Handling	Auto-expire, renewal reminders	High
F3.5	PT Session Credits	Include PT sessions in packages	Medium
F3.6	Membership Analytics	Track subscriptions, revenue	Medium

10.1.4 4. Personal Training

ID	Feature	Description	Priority
F4.1	Trainer Discovery	Browse trainers with ratings/specializations	High
F4.2	Trainer Assignment	Request and assign trainers	High
F4.3	Session Booking	Schedule PT sessions with availability	High

ID	Feature	Description	Priority
F4.4	Session Management	Complete, cancel, reschedule sessions	High
F4.5	Workout Plans	Trainers create customized plans	Medium
F4.6	Diet Plans	Nutritional guidance from trainers	Medium
F4.7	Session Ratings	Members rate completed sessions	Medium

10.1.5 5. Progress Tracking

ID	Feature	Description	Priority
F5.1	Body Measurements	Track weight, body fat, dimensions	High
F5.2	Progress Photos	Upload before/after images	Medium
F5.3	Workout Logging	Record daily workout activities	Medium
F5.4	Goal Setting	Define and track fitness goals	Medium
F5.5	Progress Charts	Visualize progress over time	Medium
F5.6	Achievement Badges	Gamification rewards	Low

10.1.6 6. Communication

ID	Feature	Description	Priority
F6.1	Real-time Chat	WebSocket-based messaging	High
F6.2	File Attachments	Share images, documents in chat	Medium
F6.3	Message Reactions	React to messages with emojis	Low
F6.4	Read Receipts	Track message delivery/read status	Low
F6.5	Notifications	In-app and push notifications	High
F6.6	Email Alerts	Configurable email notifications	Medium

10.1.7 7. Analytics & Reporting

ID	Feature	Description	Priority
F7.1	Owner Dashboard	Revenue, member stats, trends	High
F7.2	Trainer Reports	Session history, ratings, earnings	Medium
F7.3	Member Stats	Personal progress, attendance	Medium
F7.4	Financial Reports	Revenue breakdown, projections	Medium
F7.5	Export Reports	Download as PDF/CSV	Low

10.2 Non-Functional Requirements

10.2.1 1. Security

ID	Requirement	Implementation	Priority
NF1.1	Authentication	JWT tokens with expiry	Critical
NF1.2	Authorization	Role-Based Access Control (RBAC)	Critical
NF1.3	Password Security	BCrypt hashing (12 rounds)	Critical
NF1.4	Transport Security	HTTPS/TLS everywhere	Critical
NF1.5	Input Validation	Server-side validation	Critical
NF1.6	CORS Policy	Whitelist allowed origins	High
NF1.7	Rate Limiting	Prevent brute force attacks	High
NF1.8	SQL Injection Prevention	JPA Parameterized queries	Critical
NF1.9	XSS Prevention	Content sanitization	High

10.2.2 2. Performance

ID	Requirement	Target	Priority
NF2.1	API Response Time	< 200ms (95th percentile)	High
NF2.2	Page Load Time	< 3s initial, < 1s subsequent	High
NF2.3	Database Queries	Optimized with indexes	High
NF2.4	Caching	Spring Cache for frequent data	Medium

ID	Requirement	Target	Priority
NF2.5	Connection Pooling	HikariCP (max 20 connections)	High
NF2.6	Lazy Loading	JPA relationships optimized	Medium

10.2.3 3. Scalability

ID	Requirement	Description	Priority
NF3.1	Multi-Tenancy	Isolated data per gym	High
NF3.2	Stateless Auth	JWT enables horizontal scaling	High
NF3.3	Database Scaling	Supports read replicas	Medium
NF3.4	WebSocket Scaling	Dedicated connection management	Medium
NF3.5	File Storage	External storage ready	Low

10.2.4 4. Reliability

ID	Requirement	Implementation	Priority
NF4.1	Data Integrity	Database transactions (ACID)	Critical
NF4.2	Soft Deletes	is_deleted flag, recoverable data	High
NF4.3	Optimistic Locking	@Version annotation	High
NF4.4	Error Handling	Global exception handler	High
NF4.5	Audit Logging	Track critical changes	Medium
NF4.6	Backup Strategy	Database backup support	High

10.2.5 5. Usability

ID	Requirement	Description	Priority
NF5.1	Responsive Design	Works on all screen sizes	High
NF5.2	Intuitive Navigation	Clear menu structure	High
NF5.3	Loading States	Skeleton loaders, spinners	Medium
NF5.4	Error Messages	User-friendly error display	High

ID	Requirement	Description	Priority
NF5.5	Accessibility	WCAG 2.1 AA compliance	Medium
NF5.6	Internationalization	Multi-language support ready	Low

10.2.6 6. Maintainability

ID	Requirement	Description	Priority
NF6.1	Code Structure	Clean architecture layers	High
NF6.2	API Documentation	REST endpoints documented	Medium
NF6.3	Type Safety	TypeScript frontend, Java backend	High
NF6.4	Version Control	Git with branching strategy	High
NF6.5	Code Quality	Lombok for boilerplate reduction	Medium

10.3 Feature Matrix by Role

Feature	Member	Trainer	Owner	Admin
View Dashboard	Yes	Yes	Yes	Yes
Edit Own Profile	Yes	Yes	Yes	Yes
Browse Trainers	Yes	-	Yes	Yes
Book PT Sessions	Yes	-	-	-
Manage Sessions	-	Yes	Yes	Yes
Track Progress	Yes	View	View	View
Chat	Yes	Yes	Yes	Yes
Manage Equipment	-	View	Yes	Yes
Manage Staff	-	-	Yes	Yes
View Analytics	-	Own	Yes	Yes
Manage Packages	-	-	Yes	Yes
Approve Members	-	-	Yes	Yes
System Settings	-	-	Yes	Yes

Chapter 11

Chapter 7: Conclusion

11.1 7.1 Project Review and Discussion

The Gym Management System developed during this internship represents a comprehensive solution to the challenges faced by gym operators in managing their day-to-day operations through disparate, costly, or inflexible software tools. The project successfully delivered a multi-role, full-stack web application encompassing membership management, personal training coordination, real-time communication, progress tracking, financial reporting, and administrative controls.

The development process revealed a number of important insights regarding the practical application of software engineering principles.

11.1.1 7.1.1 Achievement of Objectives

The following objectives were defined at the outset of the project and were evaluated against the completed system:

Objective	Outcome
Multi-role authentication and authorisation	Achieved — JWT and RBAC implemented across all four roles
Membership lifecycle management	Achieved — complete purchase, approval, expiry, and renewal workflows
Personal training booking system	Achieved — session creation, cancellation, and rating workflows functional
Real-time chat between roles	Achieved — WebSocket STOMP messaging with delivery status

Objective	Outcome
Financial reporting and analytics	Achieved — revenue charts, transaction tables, and KPI dashboard
Security compliance with OWASP guidelines	Achieved — BCrypt, input validation, CORS, and JWT verified
Sub-200ms API response times	Achieved — validated under standard load conditions

11.1.2 7.1.2 Problems Encountered

Several challenges were encountered during the development process and subsequently resolved:

1. **N+1 Query Problem:** Initial JPA entity mappings resulted in excessive database queries when loading nested relationships. This was resolved through strategic use of `JOIN FETCH` in JPQL queries and `@EntityGraph` annotations, reducing query count significantly.
2. **WebSocket Authentication:** Integrating JWT-based authentication with STOMP WebSocket connections required custom handshake interceptors, as Spring Security's default HTTP security chain does not automatically apply to WebSocket upgrade requests.
3. **Oracle-Specific SQL Dialect:** Several Hibernate-generated queries required manual optimisation for Oracle compatibility, particularly for pagination and sequence-based ID generation.
4. **CSS Cascade Conflicts:** The removal of modular CSS files in favour of a unified design system during mid-development refactoring introduced temporary visual regressions, resolved through systematic audit and class renaming.

11.2 7.2 Novelties and Contributions

The project achieved the following notable contributions:

- **Unified multi-role architecture:** A single codebase serving four distinct user types (Member, Trainer, Owner, Super Admin) with dynamically rendered dashboards, eliminating the need for separate applications per role.
- **Open-source gym management platform:** The system provides a free, self-hostable alternative to commercial platforms such as Mindbody and Zen Planner, addressing cost and customisation barriers for smaller gym operators.
- **Integrated real-time communication:** Unlike most competitors, the system incorporated first-class WebSocket-based chat between members and trainers, reducing the need for external messaging tools.
- **Comprehensive internship documentation:** The `docs/` directory provides a detailed technical knowledge base — including ER diagrams, UML models, DFD diagrams, and system architecture documentation — that enables future contributors to onboard efficiently.

11.3 7.3 Personal Insights

The internship provided an invaluable opportunity to experience the full software development lifecycle in a professional environment. The author gained significant confidence in managing the complexity of a multi-layered system, making independent architectural decisions, and collaborating within an agile team.

Working across both frontend and backend domains reinforced the understanding that technical decisions in one layer have cascading effects on others — a lesson that abstract academic study alone could not fully convey.

11.4 7.4 Future Work

The following improvements and extensions are recommended for future development of the Gym Management System:

Enhancement	Priority	Description
Mobile Application	High	Native iOS/Android companion app for members

Enhancement	Priority	Description
Payment Gateway Integration	High	Stripe or Razorpay for online membership payments
AI-Powered Workout Recommendations	Medium	Machine learning model for personalised plans
Microservices Migration	Medium	Decompose monolith into independently scalable services
Multi-language Support (i18n)	Medium	Internationalisation for non-English markets
GDPR Compliance Module	High	Data export, anonymisation, and right-to-erasure features
Video Conferencing Integration	Low	Embedded video for remote PT sessions
Advanced Analytics Dashboard	Medium	Predictive churn analysis, retention metrics

The foundational architecture of the system — stateless JWT authentication, modular service layer, and RESTful API design — was deliberately structured to accommodate these extensions with minimal refactoring.

Chapter 12

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12.7 Citation Format

This document follows IEEE citation style. All URLs were verified as accessible as of February 2026.

12.8 Acknowledgments

- React and Spring Boot communities for comprehensive documentation
- OWASP for security guidelines

- Academic researchers in fitness technology domain

Chapter 13

Appendices

Appendix A: System Installation Guide

13.0.1 A.1 Prerequisites

The following tools must be installed before setting up the Gym Management System locally:

Tool	Minimum Version	Purpose
Node.js	18.x	Frontend runtime
npm	9.x	Package manager
Java JDK	17+	Backend runtime
Maven	3.8+	Java build tool
Oracle Database	19c+	Primary data store
Git	2.x	Version control

13.0.2 A.2 Frontend Setup

```
# Clone repository
git clone https://github.com/coded-with-aryan0426/gym-management-system-fullstack
cd gym-management-system-fullstack/frontend

# Install dependencies
npm install

# Configure environment
cp .env.example .env

# Edit .env with your API base URL
```

```
# Start development server
```

```
npm run dev
```

The frontend development server starts on `http://localhost:5173` by default.

13.0.3 A.3 Backend Setup

```
# Navigate to backend directory
```

```
cd gym-management-system-fullstack/backend
```

```
# Configure database connection in application.properties
```

```
# spring.datasource.url=jdbc:oracle:thin:@localhost:1521:orcl
```

```
# spring.datasource.username=YOUR_DB_USERNAME
```

```
# spring.datasource.password=YOUR_DB_PASSWORD
```

```
# Build and run
```

```
./mvnw spring-boot:run
```

The backend API server starts on `http://localhost:8080` by default.

13.0.4 A.4 Environment Variables Reference

Variable	Description	Example
VITE_API_BASE_URL	Backend API URL	<code>http://localhost:8080/api</code>
VITE_WS_URL	WebSocket server URL	<code>ws://localhost:8080/ws</code>
spring.datasource.url	JDBC connection string	<code>jdbc:oracle:thin:@...</code>
jwt.secret	JWT signing secret (min 256-bit)	[secure-random-string]
spring.mail.host	SMTP host for email OTP	<code>smtp.gmail.com</code>

Appendix B: Database Schema Reference

13.0.5 B.1 Core Tables Summary

Table Name	Primary Key	Description
users	user_id	All user accounts across roles
roles	role_id	Role definitions (MEMBER, TRAINER, OWNER, ADMIN)
gyms	gym_id	Gym tenant records
memberships	membership_id	Active and historical memberships
membership_packages	package_id	Configured membership packages
pt_sessions	session_id	Personal training session bookings
trainer_details	trainer_id	Trainer profile extensions
conversations	conversation_id	Chat conversation threads
messages	message_id	Individual chat messages
notifications	notification_id	System notifications

13.0.6 B.2 Key Relationships

- Each `users` record is associated with one or more `roles` via a join table.
- Each `gym` has exactly one `owner` (foreign key to `users`).
- Each `membership` references one `membership_package` and one `user`.
- Each `pt_session` references one `trainer` `user`, one `member` `user`, and belongs to one `gym`.
- Each `conversation` contains zero or more `messages`, and involves two or more `users`.

Appendix C: API Endpoints Reference

13.0.7 C.1 Authentication Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Authenticate and receive JWT
POST	/api/auth/refresh	Refresh expired JWT
POST	/api/auth/logout	Invalidate session
POST	/api/auth/verify-otp	Verify 2FA OTP

13.0.8 C.2 Membership Endpoints

Method	Endpoint	Description
GET	/api/memberships/packages	List available packages
POST	/api/memberships/request	Submit membership purchase request
PUT	/api/memberships/{id}/approve	Owner approves membership
PUT	/api/memberships/{id}/reject	Owner rejects membership
GET	/api/memberships/my	Get current member's membership

13.0.9 C.3 PT Session Endpoints

Method	Endpoint	Description
GET	/api/sessions/trainers	Browse available trainers
POST	/api/sessions/book	Book a PT session
PUT	/api/sessions/{id}/complete	Mark session as complete
PUT	/api/sessions/{id}/cancel	Cancel a booked session
POST	/api/sessions/{id}/rate	Submit session rating

Appendix D: Technology Dependencies

13.0.10 D.1 Frontend Dependencies (Selected)

Package	Version	Licence
react	18.x	MIT
react-dom	18.x	MIT
react-router-dom	6.x	MIT
axios	1.x	MIT
socket.io-client	4.x	MIT
framer-motion	10.x	MIT
recharts	2.x	MIT
lucide-react	latest	ISC
typescript	5.x	Apache-2.0
vite	5.x	MIT

13.0.11 D.2 Backend Dependencies (Selected)

Package	Version	Licence
spring-boot-starter-web	3.x	Apache-2.0
spring-boot-starter-security	3.x	Apache-2.0
spring-boot-starter-data-jpa	3.x	Apache-2.0
spring-boot-starter-websocket	3.x	Apache-2.0
jjwt-api	0.11.x	Apache-2.0
lombok	1.18.x	MIT
ojdbc8	21.x	Oracle
spring-boot-starter-mail	3.x	Apache-2.0